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Analysis of metagenomic data

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Competing interests

The authors declare no competing interests.

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Abstract

Metagenomics has revolutionized our understanding of microbial communities, offering unprecedented insights into their genetic and functional diversity across Earth's diverse ecosystems. Beyond their roles as environmental constituents, microbiomes act as symbionts, profoundly influencing the health and function of their host organisms. Given the inherent complexity of these communities and the diverse environments where they reside, the components of a metagenomics study must be carefully tailored to yield accurate results that are representative of the populations of interest. This Primer article examines the methodological advancements and current practices that have shaped the field, from initial stages of sample collection and DNA extraction to the advanced bioinformatics tools employed for data analysis, with a particular focus on the profound impact of next-generation sequencing (NGS) on the scale and accuracy of metagenomics studies. We critically assess the challenges and limitations inherent in metagenomics experimentation, available technologies and computational analysis methods. Beyond technical methodologies, we explore the application of metagenomics across various domains, including human health, agriculture and environmental monitoring. Looking ahead, we advocate for the development of more robust computational frameworks and enhanced interdisciplinary collaborations. This Primer serves as a comprehensive guide for advancing the precision and applicability of metagenomic studies, positioning them to address the complexities of microbial ecology and their broader implications for human health and environmental sustainability.

Introduction

Metagenomics is an interdisciplinary field encompassing experimental and computational methods for analyzing the genomic content and functional potential **[G]** of microbial communities. Metagenomic studies typically begin with the collection of an environmental sample of interest, such as soil, water, blood, or stool. After collection, the total DNA within the sample is extracted and sequenced using whole-genome shotgun sequencing (WGS) to generate reads originating from random genomic loci, ultimately generating a metagenomic profile **[G]** to obtain a more extensive understanding of the microorganisms in the sample. This approach contrasts with amplicon-based approaches for profiling bacteria that selectively amplify and sequence the 16S and 18S small subunit ribosomal RNA (SSU rRNA) marker genes **[G]** (see Supplementary Boxes 1 and 2).

Metagenomics originally emerged as a powerful tool for exploring the extensive microbial diversity within environmental samples such as soil¹ and water², providing unprecedented insights into the complexity and function of microbial communities in their natural habitats. As the field has evolved, metagenomics has become a foundational methodology, driving advances in microbial community research, and has been applied to critical areas such as human health, food safety, agriculture, and biotechnology. Early metagenomic methods relied on Sanger sequencing, which involves the cloning of randomly fragmented DNA into culturable bacteria³. Current methods are based on next-generation sequencing (NGS), which has dramatically reduced the cost per sequenced base by several orders of magnitude and increased accessibility for metagenomic data production (Fig. 1). Improved affordability has catalyzed the launch of numerous global-scale metagenomics initiatives^{4–7}, which have been pivotal for discovering novel microbes and enhancing our understanding of how microbes interact with their environments.

Despite the advancements made for metagenomic studies, the absence of standardized protocols for sample processing currently limits reproducibility⁸ as the quality and quantity of extracted DNA are greatly influenced by the methods used for sample collection and DNA extraction^{8–10}. Contamination poses an additional challenge, particularly in low-biomass samples¹¹ or those affected by environmental factors such as extreme climates in environmental metagenomics **[G]** studies¹². The choice of sequencing technology further constraints sample preparation^{13,14}. Additionally, the lack of consistent standards across reference databases leads to incomplete and inconsistent records^{15,16}. Enhancing experimental protocols, computational methods and analytical approaches remain active areas of research, and efforts are underway to standardize the collection and processing of diverse metagenomic samples to mitigate contamination, address issues with low-biomass samples and ensure more robust metagenomic analyses.

In this Primer, we offer an overview of essential metagenomics concepts and describe current practices for the generation and analysis of metagenomics data, the applications of metagenomics, initiatives to improve metagenomic research and the future scope of the metagenomics field. We also review the current limitations of the experimental and computational methods used in metagenomic studies and discuss possible strategies to mitigate these issues. Finally, we emphasize the need for the synergistic design of the

experimental and computational components of metagenomics experiments, as well as collaborations between basic science and applied practice, to facilitate novel applications of metagenomics in areas such as environmental monitoring, agriculture, biotechnology and medicine.

Experimentation

The choice of an appropriate sample collection method, preservation technique and sequencing workflow is necessary to form a reliable foundation for downstream metagenomic preprocessing and analysis. Given that each step can introduce bias¹⁷, the entire process from sampling through laboratory analysis should be carefully designed and executed^{18–22}. Different standardized protocols are required depending on whether microbiome [G] samples are sourced from host-associated or environmental sources. Collected samples must be handled carefully before being transported and stored diligently to maintain the integrity of metagenomic information prior to DNA extraction. After DNA extraction, library preparation steps including DNA fragmentation, end-repair, adapter ligation, and sequence indexing are required to render the sampled DNA compatible with the sequencing platform being used. We discuss the experimental protocol to prepare a metagenomics sample for bioinformatics analyses below; for an overview of the experimental workflow, see Fig. 2.

Study design

A rigorously designed study maximizes the potential for generating high-quality, reproducible data and ensures that the resulting conclusions are scientifically robust and defensible. The study design must be carefully aligned with specific research objectives, as it directly impacts the validity and interpretability of results. Researchers should select the most appropriate design based on the study's goal, which can range across human, animal, and environmental studies. Observational studies monitor a subject without intervention to reveal natural microbiome variations. Cross-sectional studies provide a snapshot of the microbiome at a single time point, which can be useful to find associations between the microbiome and specific outcomes (such as health outcomes). Case-control studies compare microbiomes between individuals with a condition and those without, identifying condition-linked differences. Longitudinal studies track microbiome changes over time, offering insights into temporal dynamics. Finally, randomized controlled trials — the gold standard for causal inference — rigorously assess the effects of specific interventions by minimizing biases. After selecting the appropriate study design, it is critical to estimate the necessary sample size that will give sufficient statistical power to detect any expected effects. This process can be achieved through a power analysis, using effect size estimates from pilot studies or existing literature^{23,24}.

Microbiome sample collection

Several factors should be considered for sample collection: whether the sample is representative of the study's objective; the potential for contamination from extraneous sources or cross-contamination; the practicality of sample collection, such as sampling costs,

convenience, and efficiency; and the nature of positive and negative controls. We consider these aspects below for a variety of different types of microbial community samples.

Human microbiome samples: The majority of human microbiome studies have analyzed the gut microbiome using fecal specimens owing to their large bacterial biomass and ease of collection^{9,10}, although many other tissue types and specimens have been investigated, including human milk²⁵, tumors²⁶, the respiratory tract²⁷, the vaginal environment,^{28,29} the urinary tract³⁰, skin³¹, and saliva³². Priorities when collecting these samples include maximizing patient comfort and compliance while also ensuring robust preservation of bacterial communities. Challenges of both in-patient and out-patient sample collection include the need for freezing methods for sample preservation, contamination, low-biomass samples and availability of hospital resources³³. In the context of fecal specimens, studies evaluating aspects of collection protocols such as the stabilization of samples and freezing methods have reported varied results on how DNA composition is impacted by different preservation methods; however, freezing freshly collected stools remains a gold standard to yield a reliable taxonomic resolution^{16,33}. Commercial products are available for at-home stool collection; self-collection of the first bowel movement of the day is usually recommended¹⁶ along with immediately freezing the sample or storing it in a preservation solution³⁴.

Non-human animal samples: In animal studies, sample collection varies depending on the study's objective, the tissue or specimen of interest and the host species. Fecal sample collection, rectal swabbing and post-mortem sampling of the intestinal contents of the host animal are options for gut microbiome studies. There is debate in the field about the interchangeability of sampling strategies. For example, in birds and reptiles, there is an ongoing debate regarding the suitability of swabbing the cloaca to sample the gut microbiome^{35,36}, given the convergence of the urinary, reproductive, and gastrointestinal tracts at that site. In mice and rats, fecal pellets and cecal contents collected at sacrifice are common samples for analysis. The non-invasive collection of fecal pellets facilitates repeatable longitudinal studies preventing bias statistical interpretations³⁷, whereas cecal contents yield a more comprehensive profile of the gut microbial community but requires euthanization³⁷. Therefore, considering the experimental question and type of study required is important when selecting the appropriate sampling method.

Environmental samples: Environmental samples for metagenomics include air samples, swabs of surfaces, and water and soil samples. Air sampling typically involves passing air through filters of various sizes to enrich for airborne microbes^{38–40}. This method is particularly challenging because of the low bacterial biomass of the resultant samples⁴¹ and because weather conditions such as wind, temperature, or humidity can influence the microbial community yielded by this method⁴². Surface samples, such as samples of frequently touched surfaces in urban regions, can be collected using flocked swabs⁶. The composition of microbial communities obtained by this method can be influenced by variables such as surface type, cleaning frequency, and human activity⁴³. By contrast, water samples should be collected at a prespecified depth and filtered upon collection^{44,45}. Soil samples should be collected using sterilized standard tools and stored in a sterile

bag. Soil sampling should consider the spatial and temporal variability of the sampled soil environment. For example, the density of microbial populations in the topsoil and subsoil differ, resulting in profile variations between these sampled communities. It is therefore important to note that microbial profiles can vary greatly across sample locations and seasons. To control for these variations, standardizing sampling protocols to sample across multiple time points and locations can capture natural microbial fluctuations; further, adjusting for environmental metadata (such as temperature, humidity or nutrient concentration) can further ensure that observed differences reflect true biological or environmental variations. Our recommendation is for researchers to choose a sampling method that minimizes the effects of contamination by maintaining a consistent sampling strategy throughout the study and to collect negative control samples, such as swabs of sampling equipment, air or areas adjacent to the sampled area, depending on the method used.

Sample handling, transport, and storage

As microbiome communities are living and metabolically active meta-organisms, samples should be frozen or inactivated immediately after collection and then maintained under stable conditions until DNA extraction. Temperature fluctuations^{34,46}, oxygen exposure, and multiple freeze-thaw cycles⁴⁶ should be avoided as these can compromise DNA and RNA integrity, affecting the metagenomic profiles of microbial communities⁴⁷. Multiple freeze-thaw cycles can also bias sequencing results by selecting for harder-to-lyse bacteria.

The gold standard method for sample handling after sterile collection is snap-freezing in liquid nitrogen, followed by storage at -80°C . However, this is often not possible in field studies and many alternative methods for sample stabilization and storage have been evaluated^{48–51}. These include preservation buffers such as RNAlaterTM, ethylenediaminetetraacetic acid (EDTA) and ethanol-based or guanidine isothiocyanate-based⁵² stabilizing agents, which are available in various commercial kits. When using stabilizing reagents, it is imperative to maintain a proper stabilizer-to-sample ratio and to thoroughly mix the stabilizer and sample. The choice of stabilizing solution, storage time and storage temperature can affect sample degradation, even for samples that were transferred immediately to -80°C . For example, the efficiency of stabilizers can fluctuate between metagenomic samples, influencing the identified taxonomic composition of a microbial community as shown in fecal specimens⁵³. Additionally, prolonged storage⁵⁴ or fluctuations in temperature⁴⁶ can potentially compromise the integrity of the microbial community and bias downstream analyses, as has been seen in pig fecal⁵⁵ and sewage samples⁵⁶. Thus, the optimization of methods used to ensure stability in metagenomic samples is important to ensure reliable interpretation of microbial communities.

To ensure reproducibility, details of preservation methods and storage temperatures should be included in study metadata when possible. Additionally, this information should be uploaded to data repositories, such as protocols.io⁵⁷ and STAR methods⁵⁸, to allow for broad use by the research community¹⁷. Freeze-thaw cycles should be avoided by aliquoting samples prior to long-term cryopreservation, and appropriate negative and positive controls should always be included for data accuracy and potential bias quantification. Although

bias due to preservation methods is unavoidable, careful consideration of sample storage and downstream bioinformatics pipelines can minimize its overall impact on the research objective.

DNA Extraction

DNA extraction (Fig. 2b) involves the lysis of microbial cells and the isolation and purification of the now-accessible DNA. Among all sample preparation steps, DNA extraction methodology might have the greatest impact on study variability due to unique biases in different microbial lysis approaches^{18,59–61}. The type of DNA extraction method used is dependent on the type of sequencing platform that will be used (short-read or long-read). A bead-beating step should be used for the extraction of samples for short-read sequencing to ensure the effective lysis of gram-positive bacteria and fungal cells, and reduce hands-on extraction time^{21,50,60–63}. Conversely, enzymatic lysis techniques should be used for samples for long-read sequencing as mechanical lysis methods such as bead-beating results in DNA shearing^{64,65}, although there are still unresolved issues with respect to enzymatically lysing fungal cells⁶⁶. Recent advancements have enabled the production of long reads from DNA extractions, which has led to the development of novel protocols⁶⁴ and benchmarking studies^{54,66,67} that take these new capabilities into account. DNA extraction kits or methods such as phenol-chloroform or CTAB extractions use different techniques to increase DNA yield and remove enzyme inhibitors from their samples. Benchmarking studies have been performed within and across different environmental samples in order to identify the best DNA extraction approaches for short-read metagenomics^{51,59,61–65}; although some studies recommend particular kits or approaches for certain types of environmental samples, no single kit or approach has been identified as best.

Incorporating negative controls is crucial for detecting contamination across sampling and DNA extraction stages, which might otherwise lead to erroneous background detection of microbes and antibiotic resistance genes [G] (ARGs)⁶⁶. Using a field blank (filling a sampling tube with molecular-grade water) effectively captures potential contamination sources within the workflow. Additionally, positive controls (such as a mock community of bacteria processed as a separate sample) or internal standards (whole cells or exogenous DNA or RNA added to the sample matrix) should be included to identify potential biases in sample concentration, DNA extraction and bioinformatics analyses.

Library preparation and sequencing

After DNA extraction, the library preparation process (Fig. 2c) converts raw DNA extracts into labeled libraries ready for sequencing on the platform of choice⁶⁷. Short-read library construction generally includes four main steps: DNA fragmentation into smaller, random, overlapping fragments using either sonication or enzymatic methods⁶⁸; fragment end-repair; ligation of pair-end or single-end adapters; and indexing⁶⁹. Indexing allows multiple samples to be combined in a single sequencing run, reducing monetary cost per sample and increasing throughput per run. Indexed libraries are normalized and pooled to equimolar concentrations prior to sequencing⁷⁰.

Sequencing platforms. Illumina NGS platforms are often used for sequencing in metagenomic studies for their high-throughput and high-accuracy generation of short reads (ranging from 50 to 300 bp). These platforms are highly parallelized, allowing for the simultaneous sequencing of many DNA fragments, and have been successfully applied to shotgun metagenomics [G] research to date⁷⁰. These platforms are ideal for tasks such as variant detection, gene expression analysis, and metagenomic profiling, where throughput and cost are paramount considerations. The established infrastructure surrounding short-read technologies, including robust analysis pipelines, further strengthens their appeal for high-throughput applications.

Long-read sequencing technologies such as Nanopore (Oxford Nanopore Technologies, ONT) and PacBio (Pacific Biosciences) can produce reads several hundred kilobases in length and have been successfully applied for metagenomics research. These approaches allow real-time data acquisition as they can generate ultralong reads that can be used to generate full-length mRNA or viral sequences. This allows direct sequencing from native DNA, reducing error rates, and allowing solving complex region sequences. Long-read sequencing excels in providing comprehensive genome assemblies, accurate mapping of complex regions and identification of large structural variants. This technology can sequence native molecules without amplification, thereby avoiding PCR-induced biases and preserving epigenetic modifications, making it valuable for specific applications such as de novo assembly and isoform identification. For example, when investigating the genetic context of ARGs and their association with mobile genetic elements (MGEs) or host organisms, long-read sequencing offers superior accuracy owing to the reads spanning larger genomic regions^{71,72}, thereby reducing the bioinformatics biases commonly introduced by short-reads⁷³. Despite these advantages, Nanopore sequencing platforms exhibit a higher average base error rate (4–10%) compared to Illumina technologies (approximately 0.1%)⁷⁴ and Illumina platforms generally outperform Nanopore sequencing in terms of read yield. Advancements in long-read sequencing technologies such as the Nanopore PromethION sequencing platform^{75,76}, the R10.4.1 flowcell and HiFi sequencing using the PacBio Sequel II system have been used to address limitations regarding sequencing depth [G], error rate, and overall sample coverage in metagenomic research⁷⁷.

When selecting a sequencing platform, researchers must consider the goals of their study, the level of genomic complexity of their samples, and a balance between cost-efficiency and data accuracy. The cost disparities between short-read and long-read sequencing can be substantial, with the cost of long-read platforms typically increasing several times per gb of data generated⁷¹. This difference arises from the longer duration of runs and greater complexity of library preparation and sequencing chemistry for long-read sequencing^{71,72}. Consequently, while long-read sequencing provides deeper insights into genomic architecture, its higher per-unit cost can be prohibitive for large-scale projects. To avoid these limitations, hybrid sequencing approaches have gained traction by integrating the accuracy and cost-efficiency of short reads with the long-range continuity of long-reads, thus offering a practical compromise and enabling high-quality genome reconstruction at a reduced cost^{73,77}.

Library preparation and sequencing. Library preparation for Illumina sequencing can be classified into PCR-based and PCR-free methods. PCR-free methods typically require a minimum DNA input of approximately 25ng, whereas PCR-based library preparation is flexible in terms of DNA input, although it can introduce biases due to the PCR amplification process. After selection and the preparation of the library, adapter-ligated DNA molecules are immobilized onto the surface of a flowcell, which is coated with oligonucleotides complementary to the adapters. These DNA molecules are then amplified through a process known as bridge amplification^{77–79} and sequenced using an optical detection method. Illumina's NGS platforms offer single-read sequencing, which sequences DNA from one end of a fragment, and paired-end sequencing, which captures both ends of a DNA fragment⁸⁰.

The effectiveness of long-read sequencing heavily depends on the library preparation step. Over-shearing during preparation can compromise read length and quality. Specialized kits are available for library preparation. In ONT's DNA-by-ligation, Rapid and 16S library prep kits, DNA is sheared to fragments longer than 8 kb, end-repaired, and ligated with protein-conjugates adapters, followed by a conditioning step before sequencing⁸¹. PacBio's SMRTbell library prep kit involves ligating universal hairpin adapters to sheared or amplified DNA fragments, preparing them for high-fidelity, long-read sequencing⁸².

One of the main advantages of long-read sequencing options such as nanopore sequencing and PacBio's single-molecule real-time (SMRT) sequencing is their ability to produce reads that span complex genomic regions. In nanopore sequencing, bases are called when DNA passes through a nanoscale protein pore, generating fluctuations in electric current from which canonical and modified nucleotides can be identified in a context-dependent manner⁸³. By comparison, SMRT sequencing has two modes: circular consensus sequencing (CSS) for highly accurate long reads (HiFi reads), and continuous long read (CLR) sequencing for long reads where half of reads exceed 50 kb in length^{84,85}. These technologies offer considerably longer read lengths than Illumina's NGS platforms, although they come with trade-offs, such as the context-dependency of base calls, which leads to a higher error rate for nanopore sequencing, and the need for repeated base calling to build a consensus, which leads to higher costs for SMRT sequencing.

A variant of paired-end reads can be generated using high-throughput chromosome conformation capture (Hi-C), a technique initially developed to study the three-dimensional organization of chromosomes in which paired reads serve to link genomic regions that are in proximity within the cell. This arrangement results in some Hi-C pairs spanning extensive genomic regions as distant DNA segments come in close proximity within the cellular milieu. The combination of shotgun sequencing and Hi-C has led to the development of metagenomic Hi-C (metaHi-C)^{86,87}, which involves shotgun extraction of genomic fragments from a microbial sample, along with a Hi-C experiment generating DNA-DNA proximity ligations between loci within the same physical cell, thus enabling the linking of contigs [G] assembled from the shotgun sequencing.

Sample multiplexing is a common practice in metagenomic data pre-processing for conducting high-throughput and cost-effective analyses of several microbial communities.

A metadata file containing details of individual indexes and their corresponding samples is essential for accurate demultiplexing. In addition, indexes should be designed based on error-correcting bioinformatics methods such as CD-HIT⁸⁸, DNAClust⁸⁹, and Shepherd⁹⁰ to mitigate misassignment of reads due to sequencing errors⁷⁹.

Results

The initial stages of metagenomics data analysis include pre-processing and quality control (QC) steps, including assessing read coverage, depth analysis in relation to a reference genome, assessing of contig completeness, and additional determination of contamination levels in assembled contigs. The bioinformatic analysis of metagenomic data then moves onto more computationally demanding tasks such as metagenomics assembly and functional analysis of microbial community, which require advanced computational infrastructure such as high-performance computing clusters [G] for efficient data processing. Here, we explore an array of strategies and bioinformatic tools commonly used for efficient data processing and metagenomic analysis, and discuss challenges and mitigation strategies to obtain reliable results effectively. Commonly used analytical tools for metagenomic analyses are presented in Supplementary Table 3.

Quality control of metagenomic data

Before quality control, a demultiplexing step is essential for distinguishing individual samples based on the unique indexes incorporated into the reads during library construction. Tools for demultiplexing include Flexbar^{91,92} and Ultraplex⁹³ have been developed among others^{94–99}.

The stochastic process of WGS on metagenomic samples can introduce biases when performing metagenomic assemblies, making it challenging to accurately reconstruct genomes within a sample. Therefore, it is important to assess the quality of the raw sequencing data after demultiplexing. Read quality is measured using the Phred quality score, provided by the sequencing platform used, which is based on the probability that a base was sequenced incorrectly. Low-quality reads — those with a Phred below 30 — often contain technical artifacts such as sequencing errors, PCR artifacts and adapter sequences, and should be eliminated using quality control and data cleanup tools, such as FastQC, PRINSEQ¹⁰⁰, Trimmomatic¹⁰¹, BBTOOLS and others¹⁰². These tools leverage both the quality information provided by sequencing instruments and databases containing collections of adapters and primer sequences to facilitate the removal of sequencing adapter sequences and investigate read sequences for anomalies, such as the over-representation of *k*-mers [G] (a sign of contamination or genomic repeats that can lead to complex metagenomic assemblies). They also assess read length, quality scores, GC content, and the number or percentage of ambiguous bases. Post-trimmed reads that fall below a pre-defined threshold length are discarded.

Alternative quality control methods are used for long sequencing reads^{103–106}. For example, the NanoFilt tool filters long sequencing reads based on common parameters such as average read quality, length, and GC content¹⁰⁴, and LongQC¹⁰³ assesses the quality of long reads by evaluating the proportion of atypical reads — known as “nonsense reads” — potentially

generated from low-quality pores. Reads derived from the host genome or from other unwanted sources of contamination must be removed; automated decontamination tools such as DeconSeq¹⁰⁵ and [KneadData](#) streamline this process.

Metagenomic assembly

Analyses can be performed directly on individual reads derived from metagenomic samples; however, the assembly of reads into individual contigs and metagenome-assembled genomes [G] (MAGs) is often preferable as it provides higher resolution (more detailed and accurate differentiation among species and strains) of the genomic content of microbial communities and enables improved functional annotation¹⁰² (Fig. 3 left side).

Contig assembly. Metagenomic assembly [G] tools largely employ extensions of assembly algorithms developed for isolated genomes that have been adapted to the uneven coverage and species/strain diversity commonly found in metagenomic data. The majority of short-read metagenomic assemblers rely on a de Bruijn graph [G] —an approach that constructs a graph composed of k -mers extracted from the reads¹⁰⁷.

Genomic repeats create ambiguities that make it difficult to identify the path corresponding to the correct reconstruction of the genome. In single genomes, repeats can be identified and even resolved by analyzing the depth of coverage within the assembly (the number of times each k -mer is seen in the reads). Short read metagenomic assemblers have been extended and adapted to long read sequencing technology. The MetaFlye assembler¹⁰⁸ computes k -mer frequencies within a local neighborhood in order to account for the varied coverage within metagenomic samples, in contrast to the long read single-genome assembler Flye¹⁰⁹. Similarly, hifiasm-meta¹¹⁰ extends the hifiasm tool¹⁰⁷ by adjusting the criteria for removing short and chimeric reads to account for the potentially low coverage of some genomes in a sample. Hifiasm-meta and hifiasm were specifically developed for the HiFi technology from PacificBiosciences. Another approach developed for the assembly of metagenomic data from high-quality long reads is metaMDBG¹¹¹, which builds the assembly around minimizers identified in reads. Minimizers¹¹² are a strategy for summarizing genomic data in a small memory footprint, thus offering speed and memory advantages over de Bruijn graph approaches.

Hybrid assembly approaches combine the precision of short reads with the extended coverage of long reads and offer a powerful solution for metagenomic assembly. In a recent study assessing ten environmental metagenomic samples and *in silico*-generated data using seven assemblers (IDBA-UD¹¹³, MEGAHIT¹¹⁴, Canu¹¹⁵, MetaFlye¹⁰⁸, Opera-MS¹¹⁶, metaSPAdes¹¹⁷, and HybridSPAdes¹¹⁸), the hybrid approaches (Opera-MS and HybridSPades) consistently outperformed others in terms of accuracy¹¹⁹. Benchmarking tools for hybrid assembly revealed that Unicycler exceeds MaSuRCA and SPAdes in producing contiguous genomes, particularly when combining Illumina and ONT data¹²⁰.

Despite advancements with hybrid assembly approaches, challenges remain in using error-prone long reads for reconstructing high-quality genomes from complex metagenomes (for example, those isolated from sludge or wastewater samples). A novel haplotype-resolved hierarchical clustering-based hybrid assembly (HCBHA) approach has been developed for

this application¹²¹. This method phases short and long reads into distinct haplotypes before assembling each bacterial genome individually, enabling the reconstruction of near-complete genomes from highly complex ecosystems.

Metagenomic samples can contain multiple strains of an organism, each differing from each other in just a few genomic locations¹²². In this case, specialized assemblers similar to tools that haplotype the human genome¹²³ are required to assemble contigs from each strain. By contrast to the human genome, which has two haplotypes, the number of variants of a genome within a sample is not known a priori and is frequently larger than two. Initially, techniques for microbial community haplotyping have been developed in the context of viral quasispecies¹²⁴ [G]. We discuss the general workflow of these strain resolution approaches in Supplementary Box 3.

We note that the multi-haplotype problem is an example of an “NP-hard”^{125,126} problem, where algorithms for solving it efficiently are unlikely to exist and all practical approaches rely on heuristics that cannot guarantee a correct answer. Such heuristics might miss strains or create a mosaic composed of combinations of haplotypes that do not actually occur in the sample being analyzed. Further challenges arise from sequencing errors and the fragmented and incomplete nature of metagenomic sequencing data. An alternative is to define haplotypes or strains only with respect to a set of conserved genes, an approach that does not require a high-quality assembly¹²⁷.

Contig binning: Contigs produced by a metagenomic assembler can originate from many different organisms and cannot be separated out *a priori*. These contigs can be organized into bins intended to represent individual organisms or taxa, relying on information that was not used during the assembly process. Contigs that have a similar coverage depth within a single sample, a similar coverage profile or a similar *k*-mer composition are usually assumed to be derived from a single genome.

Metagenomic binning [G] tools typically annotate the contigs with a set of features (such as coverage and sequence composition) and then use clustering algorithms to construct the bins. For example, MetaBat¹²⁸ uses a modified *k*-medoid approach to cluster contigs based on a distance metric that combines tetramer similarity with abundance information. CONCOCT¹²⁹ also incorporates both contig abundance and sequence composition information within the context of a Gaussian mixture model to identify clusters and bins. More recently, VAMB¹³⁰ relies on variational auto-encoders (a type of neural network) to cluster together contigs which are initially represented as points in a high-dimensional space. The embedding of contigs into this space is defined by their abundance across samples and their sequence composition. Some binning tools, such as MaxBin¹³¹ and binnacle¹³², also incorporate mate-pair information in addition to coverage and sequence composition.

Bins that appear to represent nearly complete reconstructions of genomes from a metagenomic sample are called metagenomic assembled genomes (MAGs) to differentiate them from genomes reconstructed from cultured isolates. For details on the validation of MAGs, see Supplementary Box 4.

Taxonomic characterization in metagenomics

Taxonomic classification and profiling involves identifying and estimating the relative abundance of known microbial taxa in a metagenomic sample using their sequenced genomes as references. This approach requires sequence data for the taxa of interest and differs from the metagenomic assembly approach above, which aims to discover novel microbial genomes from DNA fragments in the sample. Both taxonomic classification and profiling compare sequencing reads or contigs¹³³ against a reference database, yet they differ in approach and purpose. Taxonomic classification uses taxonomic binning to assign individual reads or contigs to specific taxa, subsequently aggregating these assignments to estimate the relative abundances of taxa present in a sample. On the other hand, taxonomic profiling reports the relative abundances of taxa by comparing the overall sequence content of a sample against reference sequences. We discuss the methodologies for taxonomic classification and profiling of metagenomes and existing tools below.

Alignment-based strategies: Alignment-based methods for taxonomic classification rely on algorithms to align sequencing reads against reference genomes (Fig. 4Bb,Bc). Modern versions of these algorithms generally employ a taxonomic binning approach to determine the potential location of each read on a reference genome, assign a score evaluating the quality of the alignment, and assign taxonomic labels to the aligned reads based on sequence similarity to the reference genome (Fig. 4Bd).

Alignment strategies using entire reference genomes represent the earliest algorithms used for taxonomic classification. An example is MEGAN, which aligns each read using BLAST and assigns them to their lowest common ancestor (LCA) — the ancestral node shared across a group of species from which the read may be derived¹³³. The LCA is used because metagenomic read data often comprises short, fragmented sequences that might not align perfectly with known reference genomes, or might align to multiple reference genomes owing to high genomic similarities between closely related species or strains. By determining the LCA shared by multiple reference sequences that match a given query sequence, the LCA approach provides a robust method for taxonomic classification, effectively handling incomplete or divergent sequences, and can also be used in alignment-free or hybrid taxonomic profiling approaches (discussed below). The algorithm arranges less-conserved species closer to the “leaves” and more-conserved species closer to the “root” of a phylogenetic tree. Although MEGAN is optimized for short reads, the taxonomic binning strategy in MEGAN has been adapted to use long reads and assembled contigs as part of the MEGAN-LR¹³⁴ tool (Supplementary Box 5).

Curated subsets of unique genes with single copy number can be used as marker genes to enhance the computational efficiency of alignment-based methods. The goal of marker gene-based approaches is to create a reference database from marker genes alone to reduce the size of the reference database and optimize resource use (Fig. 4Ba). This approach was able to reduce the RefSeq dataset from 9.8 terabytes to the MetaPhlAn dataset of 10.41 gigabytes, representing a 98.95% reduction. Marker gene-based approaches can vary widely in the number of marker genes used, ranging from a few dozen to millions; for example MetaPhyler uses 31 protein-coding marker genes chosen for their efficacy in phylogenetic

analysis, spanning 581 genera, 214 families, 99 orders, 46 classes, and 27 phyla¹³⁵. Another marker-based tool, PhyloSift, uses 37 “elite” gene families alongside with four supplementary sets encompassing 16S and 18S ribosomal RNA genes, mitochondrial gene families, eukaryote-specific gene families, and viral gene families, totaling approximately 800 gene families, with a predominant representation of viral families¹³⁶.

The reduced reference database of marker genes gives a unique representation of specific taxa. The 16S rRNA gene, which is critical for bacterial and archaeal taxonomy, is ubiquitous across prokaryotic genomes and contains conserved regions that are essential for ribosomal function, as well as variable regions that allow for discrimination between different taxa. The V4 region of the 16S rRNA gene is frequently targeted for amplicon sequencing [G] in microbial ecology studies for its high variability and species-level resolution, enabling researchers to identify microbial taxa and assess their relative abundances in environmental samples.

Some marker gene-based tools, including MetaPhlAn, align metagenomic reads to clade-specific marker genes to assess microbial relative abundances. The MetaPhlAn marker genes, which encompass both bacterial and archaeal phylogenies, were selected from over 2 million potential candidates identified from available genomes, based on their high conservation within clades and minimal similarity to genes outside those clades. MetaPhlAn compares metagenomic reads against the clade-specific marker gene reference database using nucleotide alignments, thus enabling efficient estimation of clade abundances. MetaPhlAn4, an updated version of its predecessor, introduces several enhancements, including the capability to construct custom marker gene databases from MAGs, enabling researchers to conduct more comprehensive and customized studies of microbial community diversity. Custom marker gene databases in MetaPhlAn4 are curated by combining marker genes from MAGs with those from reference genomes, thereby augmenting the accuracy and specificity of taxonomic profiling by incorporating genetic data from previously uncharacterized microbial taxa found in a metagenomic sample¹³⁷

Marker gene-based strategies, such as those that use marker genes based on operational taxonomic units (mOTUs; universally conserved single-copy protein-coding genes present across diverse taxa) have been employed to improve the resolution of taxonomic profiling of microbial communities. The use of pairwise alignments for marker genes clustering has successfully identified 10 marker genes appropriate for creating over 7700 reference mOTUs for taxonomic classification and profiling¹³⁸. Additionally, more than 20,000 reference mOTUs were recently added, identified through marker genes from approximately 150,000 MAGs¹³⁹. It should be noted that marker gene-based approaches often exhibit increased specificity (true negative identification of species) over genome-based alignment strategies at the cost of decreased sensitivity (true positive identification of species), due to their dependence on conserved nucleic acid sequences to represent taxa¹³⁸.

Alignment-free, *k*-mer-based strategies: *k*-mer-based methods for taxonomic classification generate *k*-mer profiles from metagenomic data (Fig. 4Aa), compare them against those of reference databases to identify matching *k*-mers (Fig. 4Ab) — allowing reads or contigs to be classified to a corresponding reference genome without the need for an

exact genomic index — and quantify shared k -mers to generate a taxonomic profile[G] (Fig. 4Ac). These alignment-free strategies are less computationally intensive than alignment-based methods. K -mer-based methods provide more scalable solutions by incorporating k -mer usage in techniques such as exact matching¹⁴⁰, binning¹³⁰, hashing algorithms^{141,142}, and discriminative subsets¹⁴³ (for more information about k -mer-based machine learning approaches see Supplementary Box 6).

k -mer matching has been used to map query k -mers against reference k -mers, where the LCA of the genomes containing each k -mer is used for taxonomic profiling¹⁴⁰. This approach can present challenges; for example, the k -mer matching program Kraken uses a majority vote strategy for classifying reads, which might result in suboptimal performance at lower taxonomic rank levels due to high sequence similarities and sequencing errors. The Bracken tool was developed to overcome this sensitivity issue and probabilistically reassigns reads across the taxonomic tree, improving accuracy at finer taxonomic resolutions¹⁴⁴.

k -mer-based sketching approaches provide a more computationally efficient alternative to k -mer matching. These methods condense large datasets into smaller “signatures” while preserving the similarity relationships of k -mers, thereby minimizing the computational resources required for taxonomic profiling¹⁴¹. For details on specific K -mer based sketching approaches, see Supplementary Box 7).

Hybrid strategies: Hybrid-approaches to taxonomic characterization offer a promising strategy for combining the strong precision and recall in taxonomic classification at species and genus level of alignment-based methods with the efficiency of alignment-free methods. For example, one hybrid approach¹⁴⁵ uses MinHash, a hashing technique that uses subsamples of k -mers to estimate genomic similarities, which is then coupled with alignment¹⁴⁶. This combination allows Metalign to accurately profile metagenomic datasets by estimating the containment index — the likelihood that a given reference genome is present within a sample¹⁴⁶. By reducing the size of the reference, this approach optimizes the alignment process for taxonomic profiling improving efficiency without compromising accuracy.

Short-read versus long-read taxonomic profilers: The use of short-read or long-read sequencing technologies has implications for the accuracy and resolution of taxonomic profiling. Long-read sequencing technologies can enhance taxonomic analyses by providing more comprehensive sequence information, improving taxonomic resolution and accuracy of complex microbial communities¹⁴⁷. A comprehensive benchmarking study assessing 11 taxonomic classification methods, including five designed specifically for long reads, showed that long-read classifiers generally outperformed short-read classifiers¹²⁷. Short-reads frequently produce false positives, particularly at lower abundances, and require additional filtering to achieve acceptable precision. By contrast, long-read methods such as BugSeq, MEGAN-LR and DIAMOND and the generalized method sourmash exhibit high precision and recall without requiring any filtering. Long-read methods also demonstrate superior performance for detecting species at low abundance levels; for example, these methods were able to detect species at 0.1% abundance with high precision in PacBio HiFi datasets, something that was more challenging for short-read methods.

Read quality can markedly impact the performance of long-read taxonomic profilers. Long-read datasets with a large proportion of shorter reads (< 2 kb) resulted in lower precision and worse abundance estimates¹²⁷. The study further demonstrated that long-read datasets provided superior taxonomic classification results compared to short-read datasets, particularly at finer taxonomic resolution, due to the greater contiguity of long-read assemblies. At both the species and genus levels, long-read datasets provided better detection metrics compared to short-read datasets^{71,127}.

Finally, the ability of ONT platforms to provide real-time sequencing reads, which can be used immediately for taxonomic classification without compromising precision or sensitivity due to the assembly and binning process, presents an advantage over traditional methods that require lengthy culturing and molecular diagnostics^{148,149}. This is critical for metagenomic applications in clinical settings where accurate and timely pathogen identification is critical.

Functional analysis of metagenomics

Metagenomic functional analysis refers to the process of analyzing genomic sequences from metagenomic datasets to decipher the functional capabilities and potential metabolic pathways within microbial communities in specific environments^{150,151}. Functional analysis aims to uncover the capabilities and activities of these microbes, such as the biochemical pathways they use, the enzymes they produce and their roles in ecosystem processes. The first metagenomic functional analysis was carried out by cloning the functional genes isolated from environmental DNA [G] into *E. coli* and analyzing enzymatic activity using non-computational methods¹⁵². This study was followed by others that demonstrated the presence of various microbial functions in environmental samples^{3,153,154}. Large-scale studies of the human gut microbiome such as the Human Microbiome Project¹⁵⁵ have highlighted that microbial functionalities are more conserved across cohorts than taxonomic composition. The variability in taxonomic units observed across cohorts can present a reproducibility challenge for comparative metagenomic studies, as differences in taxonomic composition can complicate the identification of consistent patterns across studies, suggesting a need to shift from taxonomic units to functional groups^{156,157}. In this section, we discuss the functional analysis process.

Gene prediction approaches. Gene prediction approaches (Fig. 3 bottom row) identify potential genes from sequencing reads or assembled contigs. The key challenge of gene prediction is discriminating whether genomic regions that resemble genes are indeed authentic protein-coding genes; in complete genomes, such regions are open reading frames [G] (ORFs), whereas in metagenomic applications the fragmented nature of the data requires the consideration of incomplete ORFs. Early alignment-based tools such as CRITICA¹⁵⁸ and Orpheus¹⁵⁹ infer gene presence by aligning query sequences to known protein databases (Fig. 3 top row); as a result, these approaches are only effective at detecting a subset of genes in a sample expressing previously characterized proteins. This limitation restricts their ability to uncover novel or unique genes due to their divergence from known sequences. More recent *ab initio* gene finders employ statistical models that are trained on known gene sequences to determine which ORFs are likely to be genes. These models analyze various features, such as *k*-mer frequencies, ORF length, GC content, and sequence motifs

to predict genes. Other methods such as hidden Markov models (HMMs)^{160–165}, support vector machines (SVMs)¹⁶⁶, and conditional random fields¹⁶⁷ have also been employed in microbial gene finders to improve the accuracy of gene prediction.

The parameters of *ab initio* gene prediction models can vary across bacterial species and complicate gene prediction in metagenomic samples as the organism to which each ORF belongs is typically unknown. Tools such as MetaGene and its successor, MetaGeneAnnotator¹⁶⁸, address this issue by focusing on species-specific features such as patterns of ribosomal binding sites and di-codon usage in training their probabilistic models. This enhances the precision of gene prediction across metagenomic data¹⁶⁸. Prodigal, another widely used tool for microbial gene prediction, leverages start codon and ribosomal binding site motifs, GC content, and hexamer usage patterns to build a model that can identify and annotate genes in microbial genomes¹⁶⁹. Recently, deep learning tools like CNN-MGP¹⁷⁰ and Meta-MFDL¹⁷¹, have been used for gene finding and demonstrated promising results.

Despite the numerous gene prediction tools available, no consensus exists on a definitive gold standard. A benchmark study demonstrated that *ab initio* methods generally surpass evidence-based tools, with performances varying across different organisms¹⁷². To further optimize gene prediction, the annotation tool MetaErg incorporates additional features such as signal peptides and transmembrane helices, which are implicitly contained in trained models, to enhance the filtering of predicted ORFs¹⁷³.

Annotating functions in a metagenome. Functional annotation in metagenomics (Fig. 3 left side) involves computationally assigning biological or functional information to metagenomic sequences, facilitating an understanding of the roles and activities of microbial communities within their environments. It essentially involves identifying and elucidating the functional potential of each gene, thereby expanding the knowledge base of gene functions and potentially discovering new genes or functions. Functional annotation complements functional profiling (covered in the next section) as it focuses not only on identifying known functions but also on predicting roles for previously unknown genes, whereas functional profiling quantifies the abundance of known genes in a sample (Fig. 3 middle row)^{173–179}.

Comparing unknown genes against reference databases is a fundamental step in functional analysis. Orthology-based approaches such as eggNOG-mapper¹⁸⁰ and KEGG KOALAtool¹⁸¹ leverage databases such as Clusters of Orthologous Groups (COGs)¹⁸² and Kyoto Encyclopedia of Genes and Genomes (KEGG)¹⁸³, respectively, to classify genes into evolutionarily related groups to facilitate gene function prediction using shared ancestry. Additionally, Operational Functional Units (OFUs), which group genes or gene products based on their functional similarities, are emerging as a new benchmark for understanding the functional potential of microbiomes¹⁸⁴. Other annotation tools that use structural similarity and incorporate clustering or transfer learning approaches include MorF¹⁸⁵ and FunFams¹⁸⁶. The tools ProtBERT¹⁸⁷ and ProSE¹⁸⁸ rely on sequence embeddings. Additional efforts have been made in the construction of protein families through databases,

such as Pfam¹⁸⁹ and AGNOSTOS-DB¹⁹⁰, which are instrumental in identifying proteins with related functions and elucidating unknown boundaries of functions.

Despite advancements in functional annotation tools, a significant fraction of microbial functions can not be properly annotated¹⁷⁶. To increase the proportion of sequences with identified functions, it is important to adopt gene context-based methods beyond traditional orthology-based approaches such as FunGeCO and integrate the genomic environment surrounding a gene, thereby enhancing functional annotation¹⁸⁴. Alternatively, the subsystems approach, exemplified by MG-RAST¹⁹¹, organizes gene families into subsystems within functional networks, thus enhancing our understanding of microbial metabolism. Moreover, deep-learning based methods, such as DeepFRI¹⁹², represent a competitive alternative, effectively combining sequence with structural features to substantially increase the coverage of functional information.

Functional annotation of metagenome sequences can serve as a valuable tool to predict and discover novel genes involved in metabolic pathways and to enzymatic functions, antibiotic resistance genes, and CRISPR-CAS systems, valuable in different biotechnological applications¹⁹³. Additionally, functional annotation enables the exploration of metagenomic functional signatures across various environments, such as nitrogen cycles¹⁹⁴, disease-specific microbiomes¹⁹⁵, and virus-host microbiome interactions¹⁹⁶.

Functionally profiling metagenomic data. Functional profiling elucidates the functions encoded by the annotated genes to understand the functional dynamics and ecological roles of the diverse microbiota. This approach involves aligning query sequences against databases of sequences with known functions or employing non-alignment approaches, such as machine learning models trained on sequences with known functions. Some approaches operate directly on reads, whereas others are applied to the sequence of genes identified in the samples. The latter approach offers advantages as the analysis is performed on longer DNA segments, which reduces redundancy and improves analytic accuracy.

Alignment-based functional profiling methods use sequence comparison algorithms to compare the predicted amino acid sequences of genes with sequences in a reference database. For example, MG-RAST¹⁹⁷ uses BLAST to search against the M5nr database¹⁹⁸, a curated collection of non-redundant protein sequences from the National Center for Biotechnology Information (NCBI), UniProt, and KEGG. Additionally, BlastKOALA and GhostKOALA, which have been developed specifically for the functional annotation of protein sequences in microbial genomes, perform searches against KEGG Orthology (KO) terms¹⁸¹. DIAMOND¹⁹⁹, which is specifically designed for short reads, uses a seed search approach and double indexing to improve speed and accuracy when querying against databases such as NCBI and KEGG. Sequence alignments can enhance the capabilities of alignment-free-based profilers; for example, EggNOG-mapper¹⁸⁰, which uses pre-computed sequence clusters to facilitate comparisons, uses DIAMOND as one of its efficient sequence mapping options.

Tools using hidden Markov models (HMM), such as InterProScan²⁰⁰, have been developed to facilitate comparisons of genes against orthologies using pre-computed sequence clusters, which is less computationally intensive than searching through all sequences in a database as in alignment-based strategies. For example, KOfamKOALA²⁰¹ reduces the computational time of alignment-based programs by using HMMER/hmmbuild to produce profile HMMs and implementing an adaptive score threshold to accurately delineate metabolic and regulatory functions.

Protein domains or motifs can be annotated to further evaluate the functional properties of microbial communities. Functional annotation can use protein domains to identify conserved regions and infer putative biological functions at a proteomic level, where proteins sharing similar domains are classified into protein families, providing evolutionary insights into microbial communities. Proteomic information can be found in databases that store validated protein domains and families, such as Pfam¹⁸⁹, UnitProt²⁰² and SWISS-PROT²⁰³; tools such as HMMER²⁰⁴ and InterProScan²⁰⁰ can be used to search for domains to assign putative functions to protein sequences.

Pathway-based methods enhance functional analysis by annotating the pathways within metagenomic data through the identification and interpretation of gene functions, as well as protein and molecule actions. Such information can be particularly valuable because pathways might span multiple organisms, be partially represented, or even serve as the foundation for constructing mechanistic models. These methods, which rely extensively on well-curated reference databases, not only improve our understanding of metabolic functions but also underscore the complex interplay within microbial ecosystems. For example, HUMAnN (HMP Unified Metabolic Analysis Network) annotates and reconstructs human metabolic pathways²⁰⁵ by mapping sequencing reads to reference genes or protein families that constitute known functional pathways, enabling the exploration of ecological and biogeochemical functions between the host and the environment. Newly developed tools such as gutSMASH²⁰⁶ profile known and predicted novel metabolic pathways by using metabolic gene clusters to conduct functional comparisons in different cohorts. Pathway-based databases include MetaCyc²⁰⁷, KEGG, KBase²⁰⁸, and DrugBank²⁰⁹.

Applications

Potential applications of metagenomics are broad, spanning from clinical applications such as pathogen surveillance and monitoring antibiotic resistance and applications in ecology such as studying the crucial ecological roles of microorganisms, analyzing biodiversity and assessing ecosystem functions and the impact of environmental changes. We discuss these applications below.

Microbiome–disease associations.

Bioinformatic analysis of metagenomic sequencing data has become an important tool in understanding shifts in human microbiome composition associated with various health and disease states. This approach enables the precise identification of specific microbial taxa and functions linked to health conditions and the exploration of the complex dynamics of microbial interactions during disease progression.

Researchers often employ either a cross-sectional or a longitudinal study design to understand the association between diseases and the microbiome. In a cross-sectional study design, a cohort with a specific disease is compared with a healthy control group. Characteristics of the microbiota, such as taxonomic composition and functional profiles [G], are analyzed and compared between groups to identify differences that might correlate with health status. Typically, the relative abundance of different taxonomic groups is evaluated to determine if specific taxa correlate with disease status and statistical tests, and standard t-tests and complex linear models are used to assess associations or differences between groups²¹⁰. These techniques have been used successfully to find associations between the microbiome and both diseases of the digestive system and systemic disorders²¹⁰. For example, a decrease in butyrate-producing bacteria and an increase in functional groups responsible for sulfate reduction and oxidative stress resistance is seen in patients with type 2 diabetes²¹¹. Similarly, in major depressive disorder, a number of pathways involved in amino acid metabolism were shown to be disrupted in the fecal microbiome²¹². In rheumatoid arthritis, both the oral microbiome and gut microbiome are altered, with functional differences in the redox environment and the transport and metabolism of iron, sulfur, zinc, and arginine²¹³.

Longitudinal study designs involve sequencing the microbiota at various time points during the progression of the disease to track changes in microbial composition over time and obtain insightful information into microbiome dynamics changes alongside disease progression. For example, longitudinal studies on the oral, lung, and gut microbiota of patients with acute respiratory failure and lower respiratory tract infections performed across different hospitals have revealed distinct patterns of taxonomic and functional profiles; these studies then evaluated changes in these patterns in response to disease progression and treatment^{214,215}.

Measuring microbial diversity can aid in revealing compositional differences in microbiota across different environments and samples. Statistical measures, such as Shannon's diversity index²¹⁶ or bioinformatics tools such as PhyloSeq²¹⁷ have been developed to study microbial diversity in metagenomic samples. For example, a study by Yin and colleagues used the Shannon Index to evaluate the impact of gut microbiome diversity on mortality risk in septic shock patients²¹⁶. Using these statistical methods, researchers have identified correlations between low oral microbiome diversity and periodontitis²¹⁸, as well as between low gut microbiome diversity and various ailments, ranging from prediabetes²¹⁹ to Crohn's disease²²⁰. The practical application of microbial diversity is often limited to conditions where the microbiome's impact is well understood and the use of alpha diversity indices^{221,222} [G] to study the impact of the microbiome on health conditions such as Parkinson's, multiple sclerosis, and certain forms of depression have proved inconclusive results.

Despite significant taxonomic variability, the human gut microbiome often maintains a conserved functional profile²²³. Analyzing metagenomic data at the gene and pathway levels can help researchers elucidate the mechanisms through which the microbiome interacts with host factors in disease processes. This analysis holds promise for clinical applications in both diagnosis and treatment. For example, the recent development of [q2-predict-dysbiosis](#),

a metagenomics-based gut health evaluation tool described in a recent pre-print article²²⁴, leverages microbial functions and interactions and has shown significant improvements over most traditional taxonomy-based indices for the diagnosis of specific health conditions such as inflammatory bowel disease.

Functional analysis relies on the completeness and accuracy of functional units, such as genes, gene clusters, or protein families, which require proper categorization and distinction within samples by utilizing information on genes and pathways, as well as assessing their abundance levels. Bioinformatic tools such as Anvi'o²²⁵ or Picard can be used for this purpose for clustering contigs based on sequence similarity and annotating genes and metabolic pathways using reference-based alignment.

Clinical diagnosis using metagenomics.

Conventional diagnostic techniques rely on culturing pathogens in selective media or detecting pathogen-associated biomarkers and are therefore effective at identifying infections where suitable diagnostics are available. However, infections and their etiologies are often complex, polymicrobial or of unknown origin^{226,227}. Clinical metagenomics can offer a more comprehensive solution to traditional methods by effectively unveiling the complex microbial composition of the human microbiome — particularly valuable for polymicrobial infections that are often severe and challenging to detect and treat using conventional methods^{228–230}.

Metagenomic data can be used for monitoring antibiotic resistance^{231,232}, identification and surveillance of pathogenic strains²³³, tracking in-host microbiome evolution²³⁴, biomarker discovery²³⁵ and detecting virulence factors and toxins of different environments in patients²³⁶. Information gathered from these strategies hold incredible potential for personalized medicine^{137,236–238}. For example, clinicians can tailor treatment strategies more precisely by analyzing the resistome [G] of a clinical sample, reducing the risk of promoting antibiotic resistance²³⁹. Similarly, the detection of genes encoding toxins such as colibactin in complex biological samples from patient samples can serve as an early indicator of disease risk, such as colorectal cancer^{240,241}. By integrating such detailed diagnostic information with insights into the complex microbial ecosystem of each infection, clinicians can more accurately predict clinical outcomes and tailor treatment strategies to individual needs^{228,229,242–245}.

Although still in its infancy, metagenomics has proven to be highly effective in early pathogen diagnosis, offering a short detection time and a high sensitivity²⁴⁶. It shows improved clinical outcomes in diagnosing culture-negative sepsis and difficult-to-diagnose infections such as pneumonia, severe diarrhea, and meningitis^{247–250}. Clinical metagenomics has emerged as an important tool in understanding the associations between microbial composition and various pathologies, such as cystic fibrosis, inflammatory bowel disease, and colorectal cancer^{251–253}. Additionally, metagenomic strain profiling workflows can improve clinical diagnostics by providing tools to track strain-level variability within complex metagenomic datasets^{239,254}. For example, metagenomic strain profiling is invaluable for monitoring engraftment and tailoring treatment strategies to individual needs in the context of fecal microbiota transplant²³⁹. Furthermore, metagenomics has a crucial

role in identifying emerging and novel pathogens, a capability that has become increasingly important following the COVID-19 pandemic^{255–257}.

Wide-scale implementation of metagenomics into clinical practice has lagged behind basic research applications, largely due to challenges associated with standardization, reproducibility, cost, slow turnaround time and regulations²⁵⁸. However, the declining costs, advancements of next-generation sequencing technologies, and the emergence of innovative computational tools can pave the way for advancing the field of clinical metagenomics. Standardization and validation of practices across clinical centers continues to present a significant challenge. In particular, standard operating procedures, choice of DNA extraction methods, sample handling, and computational analyses require high levels of standardization in clinical metagenomics. Moreover, divergent practices and protocols can adversely impact the interpretation of clinical metagenomic data influencing subsequent diagnostics.

Tracking the spread of disease and surveillance of pathogens

Traditional pathogen surveillance methods strategies relying on cultured isolates face challenges related to resource demands, including the need for extensive laboratory materials, infrastructure, skilled personnel and the time to obtain pure isolates. Metagenomics offers a promising alternative to these methods by enabling culture-free detection of known and previously undetected viruses, bacteria and fungi, and can also identify virulence and resistance determinants in microbiome samples²⁵⁷. As metagenomics techniques continue to advance, they hold the potential to identify and track threats to the health of humans and farm animals as part of a comprehensive One Health approach [G]²⁵⁸.

Strain tracking, which involves the continuous monitoring of microbial strains over time and at different locations, has provided evidence of vertical transmission of microbes from maternal breast milk to the infant gut microbiome²⁵⁹. Strain tracking holds important clinical applications in the context of outbreaks of antibiotic-resistant pathogens in hospital settings. Metagenomics and other NGS approaches such as targeted whole-genome sequencing can be used to trace the spread of individual strains between patients, allowing healthcare facilities to respond appropriately with isolation measures and specific antibiotics interventions²⁶⁰. Bioinformatic pipelines, such as MIDAS²⁶¹, StrainPhlAn²⁶² and SameStr²⁶³, have been developed specifically for strain tracking and population genomics across multiple samples. These tools are tailored for the taxonomic classification of reads from similar genomes and for detecting microbes of lower abundance where traditional taxonomic assignment methods might fail to differentiate sequences at the strain level.

Metagenomics can be used to monitor microbial communities for the presence of known and emerging pathogens by systematically analyzing samples from diverse environments such as water, soil, sewage and air, facilitating the establishment of baseline microbial diversity data that can serve as reference for the surveillance of microbial temporal trends. Notable examples and initiatives of pathogen surveillance are discussed in Supplementary Box 8. Eventually, metagenomics could be used for developing and implementing metagenomic-backed policies and acting as an early warning system for the emergence of new pathogens or antimicrobial resistance [G] (AMR) for improved public health outcomes^{56,264,265}.

Environmental health monitoring and conservation.

Metagenomics can offer powerful tools for identifying microbial communities involved in processes such as nutrient cycling^{266,267}, pollutant degradation and bioremediation²⁶⁸. Soil microbiomes, comprising bacteria and fungi, facilitate nutrient cycling by decomposing organic matter and releasing essential nutrients like nitrogen and phosphorus, with diazotrophic bacteria (such *Rhizobium*) enhancing soil fertility by converting atmospheric nitrogen into bioavailable ammonia²⁶⁹. Additionally, specific bacteria release organic acids, solubilizing phosphorus compounds to aid plant uptake, a critical process for agricultural productivity and soil health²⁷⁰. In pollutant degradation, bacteria such as *Pseudomonas* secrete extracellular enzymes and glycoconjugates that enhance the breakdown of organic pollutants, metabolizing hydrocarbons in contaminated soils into simpler, non-toxic compounds through oxidation, a process essential for mitigating the long-term accumulation of industrial pollutants²⁷¹. Marine microalgae, such as *Chlorella* and *Spirulina*, play a significant role in bioremediation by biosorbing heavy metals and pollutants from wastewater, offering a sustainable solution for environmental cleanup^{272,273}. Identifying microbial communities capable of degrading pollutants and elucidating the key degradation pathways would assist in the development of tailored bioremediation strategies to enhance our ability to remove a diverse array of environmental contaminants and perform real-time monitoring of microbial community dynamics during bioremediation processes^{274,275}. Metagenomics can also be used to assess the health and biodiversity of ecosystems^{276,277}, for example by monitoring microbial community changes to determine the impacts of human activity, climate change and other environmental factors²⁷⁸. Microbial communities play a crucial role in supporting the health and resilience of animals, insects, and plants, and metagenomics may be applied to support the conservation of endangered species²⁷⁹.

Environmental metagenomics can be used for disease surveillance and outbreak prediction by monitoring microbial diversity in places such as hospitals, water plants, and farms, allowing for early intervention²⁵⁷. It could also be used for the discovery of new antibiotics from diverse environments, which is crucial for battling antibiotic-resistant bacteria^{280,281}. Metagenomics studies of built environments can provide insights into indoor air quality and pathogen dynamics²⁸².

Sustainable agriculture.

Understanding the soil microbiome is essential for sustainable agriculture and managing soil health²⁸³. Potential applications of metagenomics in this space could include investigating how plants interact with soil microbes to design microbial-based solutions for enhancing crop productivity, improving nutrient cycling, and boosting disease resistance^{284,285}. Metagenomics can also aid in the identification of soil microorganisms that are beneficial to crop species, in order to facilitate the development of biofertilizers and microbial inoculants and reduce chemical fertilizer use^{286,287}. Soil metagenomics has been invaluable for studying the rhizosphere microbiome — the microbial community surrounding plant roots — helping to pinpoint beneficial microorganisms that suppress plant diseases to promote sustainable agricultural practices. See more about soil metagenomics and metagenomics in agriculture in Box 1.

Reproducibility and data deposition

Following the first commercial high-throughput sequencer introduced in 2005²⁸⁸, the cost of high-throughput sequencing has greatly decreased, leading to a rapid increase in the amount of metagenomic data that is being generated and stored in public repositories²⁸⁹ (Table 2, Supplementary Table 4). Such repositories serve as a vital resource by facilitating the dissemination of a diverse array of publicly available data to the wider scientific community, including metagenomic sequencing data, sequence annotations, geospatial data, and computational resources (see more about public repositories for raw metagenomics data in Supplementary Box 9). The reliability and robustness of metagenomic analyses fundamentally depend on the completeness and accuracy of reference sequences from genome databases (Table 1). Existing databases cover a range of different taxa and exhibit varying levels of completeness and annotation quality in genome assemblies.

For new users of metagenomics techniques, understanding what metadata needs to be considered is crucial for ensuring comprehensive and standardized reuse of existing data from public repositories. Metadata includes various information such as sequencing parameters, sample and collection, quality control measures^{290,291}, which have been summarized in Supplementary Table 5.

The Genomic Standards Consortium²⁹², BioProject, and BioSample project²⁹³ are collectively focused on establishing standardized protocols and minimum information standards, streamline data organization, and enhance metadata quality. By promoting uniform data organization and improving metadata quality, these groups aim to ensure that datasets are more accessible and comparable across studies.

Microbial reference genome resources

Microbial reference databases might include both complete and draft genomes⁷⁸. However, specialized reference genome databases often implement rigorous quality control procedures to ensure that only assemblies meeting high standards of sequence and annotation quality are included. For example, the RefSeq database excludes assemblies derived from environmental samples owing to concerns about the accuracy of organism assignment and potential cross-contamination. Differences between reference sequence sources across databases are due to minor genomic variations among organisms and their genome organization, potentially leading to inconsistencies. Additionally, different databases can show conflicting taxonomic labels for identical species or strains¹⁵. These and other discrepancies can exist among commonly used reference sequence databases such as RefSeq²⁹⁴, Ensemble²⁹⁵ and the Pathosystems Resource Integration Center (PATRIC)²⁹⁶. Some species are uniquely represented in only one database, leading to a difference in genomic quality that cannot be reconciled by using another¹⁵. Inconsistencies and discrepancies between sequence databases can affect the outcomes of bioinformatic analyses, limiting both the accuracy and reproducibility of metagenomic studies¹⁵. Researchers are therefore urged to meticulously evaluate both the quality of raw metagenomics data and the accompanying metadata to guarantee robustness and reliability in their analyses. We suggest users consider the latest and most complete databases, such as those with regular updates such as RefSeq, GTDB, remain consistent with data

analysis and avoid switching databases unless for a good reason; and remain cautious when comparing results generated from distinct database by double-checking differences in genomic contents. A master database incorporating multiple resources, which is a complex and time-consuming task, would alleviate these issues.

The frequency and rigor of updates vary greatly across different platforms. GenBank²⁹⁷, for example, relies on original submitters to update their sequence submissions. GenBank also has less strict curation policies than other databases, primarily focusing on adding new uploads rather than curating existing data. By contrast, the RefSeq database²⁹⁸ is regularly updated to refine annotations and integrate new information, adhering to specific quality standards²⁹⁸.

The scope of organisms present within reference databases varies widely. Some databases, such as RefSeq, cover a broad range of organisms across all living kingdoms, including bacteria, viruses, archaea, and eukaryotes. By contrast, other databases narrow their focus on specific groups of species; for example, FungiDB²⁹⁹ and Joint Genome Institute (JGI) MycoCosm^{300,301} specialize in fungal reference sequences and the PATRIC database²⁹⁶ (part of BV-BRC³⁰²) is dedicated to bacterial and viral genomes.

The usability of databases for downloading reference genomes can vary. Most databases provide easy access and bulk download to all their stored genomes and other related data through public access file transfer protocol (FTP) (Supplementary Table 4). However, others, including JGI MycoCosm^{300,301}, MBGD³⁰³, and the Human Reference Gut Microbiome (HRGM)³⁰⁴ require a user registration step.

The continuous refinement and expansion of reference genome databases plays a crucial role in advancing the field of metagenomics and its applications in understanding microbial communities. Despite ongoing efforts to compile comprehensive reference sequence databases, a large portion of metagenomic reads in a sample remain unexplained, failing to be assigned to any known taxa^{6,305,306}.

Reproducibility and reusability of public metagenomic data.

High-quality public metagenomic datasets can facilitate the reuse of previously published metagenomic data, supporting open science principles that promote transparency and collaboration within the research community³⁰⁷. Easy-to-use formats and complete metadata are essential for maximizing the usefulness of these datasets. However, concerns regarding data quality, metadata incompleteness and the heavy computational requirements of analyzing raw metagenomic data have resulted in the underuse of available datasets by the scientific community^{308,309}.

One way to reuse metagenomic data is through meta-analyses^{290,310}. The aggregation of multiple studies into large meta-analyses has become a powerful practice for uncovering novel insights across diverse domains such as human health³¹¹, microbial ecology³¹², virology³¹³ and the identification of promising drug targets^{314,315}. Effectively performing meta-analyses of existing metagenomic studies is subject to numerous challenges associated

with data reuse, such as disparities in data generation, metadata incompleteness and limited public access to relevant raw data²⁹⁰.

Some recent cloud-based repositories provide integrated computational resources and workflows to facilitate the analysis of large datasets. Platforms such as MGnify³¹⁶ offer automatic analyses through established pipelines and provide processed results for immediate research use. Similarly, the NCBI's Cloud Resources and the Galaxy Project provide cloud-based access to computational tools and databases, enabling complex bioinformatic tasks without the need for high-end local infrastructure. The Galaxy Project further simplifies the process by offering a user-friendly, web-based interface that does not require programming skills. CyVerse³¹⁷ provides integrated cyberinfrastructure to support data storage and processing across both public and private clouds. Cloud-based repositories will be crucial for democratizing access to metagenomic data analysis, allowing more researchers to leverage advanced computational tools and infrastructure. However, scaling such resources to handle efficient analyses of thousands of samples remains a challenge. Further, the associated costs of these resources could be burdensome, particularly for researchers from low-resource universities and countries.

Limitations and optimizations

Several factors must be considered to counteract systemic biases and mitigate confounding factors within the study design phase of metagenomic analyses. This section outlines specific limitations and confounding factors that can be present in metagenomics analysis and discusses strategies for how they might be addressed.

Experimental and sampling biases

Sampling from different geographic locations or under varying environmental conditions can confound results due to differences in microbial populations³¹⁸. Environmental variables such as pH, temperature, moisture, and nutrient availability play important roles in shaping microbial community structures. Further, differences in the way samples are collected³¹⁹, the duration of sample storage³²⁰ and the storage medium used³²¹ can further contribute to experimental biases. Failing to consider and properly curate the metadata might distort the relationship between microbial composition and the variables of interest.

Behavioral aspects can exert considerable influence on the microbial composition of human or animal microbiomes. For example, dietary habits have been shown to shape the dynamics of the human gut microbiome, with disruption of the microbiome linked to chronic conditions such as cardiometabolic diseases and type 2 diabetes (T2D)^{156,322,323}. Failure to control for behavioural factors such as diet and lifestyle could confound study results³²⁴. Host genetics can also affect the diversity and abundance of specific taxa in the host-associated microbiome^{325,326}. Failure to account for genetic associations with the microbiome could confound the interpretation of results, obscuring the true reciprocal relationship between hosts and their associated microbiota. Metagenomic-Wide Association Studies[G] (MWAS) have been used to account for genetic associations; for example, applying MWAS on the tongue dorsum and the saliva has revealed five genetic loci associations to the oral microbiome³²⁷.

In longitudinal studies, samples analyzed after extended storage periods might differ from those sequenced closer to their sampling dates and researchers must cautiously interpret minor but consistent declines in microbial richness over time³²⁰. Adherence to rigorously standardized protocols is important to mitigate sampling and storage biases and ensure consistency and reliability of the data. Strategies to address storage issues include collecting biological and technical replicates, using standardized collection devices and procedures and maintaining aseptic conditions throughout storage and analysis to prevent contamination. Rapid freezing of samples at -80°C is also recommended for optimal preservation of sample integrity. When freezing is not feasible, the consistent use of preservatives is crucial. The duration of storage must be documented and considered during analysis to ensure accurate interpretation of results.

Three primary sources can introduce bias in metagenomic analyses during sample preparation: variable DNA extraction, contaminant DNA introduction, and inclusion of DNA from non-viable host cells or microorganisms. The use of different DNA extraction methods can result in microbial community profiles that are distinct from those present in the original samples. For example, some microbes with resilient cell walls are inadequately represented in some metagenomic datasets owing to inefficient cell lysis. Moreover, different extraction methods might yield varying amounts of DNA or exhibit biases toward certain types of microorganisms. Consequently, DNA extraction represents one of the most biased steps in metagenomic data generation^{328,329}. When managing DNA extraction biases, it is essential to employ validated kits or protocols while ensuring consistency across samples by following a standardized protocol. Documentation of extraction batches and inclusion of this information as a covariate in downstream analysis are also important for mitigating batch effects.

Contamination from the sampling and laboratory environments, equipment, reagents, and even the researchers themselves can introduce foreign DNA sequences into metagenomic data, complicating the interpretation of results³³⁰. Addressing contamination requires the implementation of rigorous quality control measures to identify potential sources of contamination and minimize its effects. Contamination from external sources might originate from kits and reagents that commonly contain small amounts of contaminating bacteria, which can vary across manufacturers and facilities^{330,331}. Additionally, cross-contamination can occur due to the inadvertent transfer of sample DNA, indexes (barcodes), and amplicons between samples, further exacerbating this issue. Although low-biomass samples are more susceptible to the deleterious effects of contaminants, any sample can be affected^{330–332}. Implementing appropriate negative controls such as sampling blanks, DNA extraction blanks and no-template amplification is recommended to address metagenomic contamination, particularly for low-biomass samples. Through such measures, contaminants can be distinguished from endogenous (native) taxa and subsequently removed from the data^{333,334}. The RIDE checklist offers valuable recommendations to reduce metagenomic contamination³³⁵, emphasizing the importance of reporting experimental design, incorporating negative controls, determining the level of contamination, and evaluating the influence that contaminant taxa have on the interpretation of metagenomic data³³⁵. Contamination can be further addressed computationally via algorithmic models

that are able to statistically detect metagenomic contamination via analysis of read frequency³³³.

Some metagenomic samples contain high levels of host DNA with low microbial biomass. Such samples include human and animal milk, which can contain up to 95% host DNA^{32,336}, host tissues, saliva, and other bodily fluids. An abundance of host DNA presents challenges for distinguishing between host and microbial reads, particularly for novel microbes. Notably, the plant holobiont is susceptible to contamination from sources such as soil microbes, co-amplified plant organelles and human DNA. These contaminants necessitate rigorous sampling protocols, specialized preparatory steps, and robust decontamination during bioinformatics analysis. Plant sample preparation should involve washing with sterilized solutes to remove potential contaminants; sequencing these reagents separately helps identify contaminants, preventing their interference with microbiome analyses and ensuring the detection of low-abundance community members^{329,337}. Common sources of host contamination in plant metagenomics include co-extraction of chloroplast and mitochondrial DNA during milling and physicochemical lysis, as well as human DNA and relic DNA from the rhizosphere. These contaminants can obscure estimates of soil microbial diversity, impacting the analysis of root samples (rhizosphere soil) and other plant tissues^{338–340}. Various strategies have been developed to address the challenges presented by host DNA including selective extraction to remove host cells^{341,342}, post-extraction methods that enrich for microbial genomes^{32,343}, and bioinformatic approaches to filter out host-associated reads post-sequencing. It is important to note that host depletion approaches might introduce bias in the sequencing of specific elements within a metagenome^{336,342}.

Low microbial concentrations: Microbial concentrations can vary across metagenomic samples, posing a challenge for metagenomic research. This issue is particularly relevant in clinical metagenomics as some body sites, such as the urine³⁴², skin³¹, lungs³⁴³, and blood³²⁸, contain low microbial concentrations. Therefore, the volume necessary for obtaining sufficient microbial biomass from non-stool samples should be considered^{344,345}. For example, a study on DNA yield from different environmental samples (soil, water and air) revealed that air samples require a much larger volume of sample to yield 5ng of DNA when compared to soil and water³⁴⁴. Alternative approaches have been presented to study the microbiomes of ultra-low biomass surfaces and the use of negative controls should be able to detect genomic contamination that can result from the use of DNA-based reagents³⁴⁶. If appropriate measures (such as sample volume³⁴⁴, reagent use^{330,346}, and the incorporation of negative controls³⁴⁶) are not taken, metagenomic analysis of low-biomass samples can lead to biased interpretations of study results.

The limit of detection for sequencing-based assays is an important consideration when using metagenomic analysis tools for clinical applications, even in samples with high bulk microbial concentrations. For example, *Clostridioides difficile* is an important human pathogen that has a relative abundance of less than 3% of the total microbial populations in stool of patients with active *C. difficile* infection (CDI), making diagnosis challenging³⁴⁷. Clinical diagnostic protocols rely on enrichment techniques, such as ex-vivo culture or PCR, or focus on highly abundant expressed genes known in CDI³⁴⁷. Untargeted

metagenomic sequencing would require extreme deep sequencing efforts to capture hard-to-detect pathogens such as *C. difficile*. Achieving clinically meaningful interpretations of metagenomic data is essential for high fidelity and requires high-quality sequences that accurately represent the microbial community present in the sample. The accuracy from high fidelity data ensures the ability to precisely predict antibiotic susceptibility. For example, unreliable data can lead to misidentification of resistance genes compromising treatment options. Therefore, the purpose in refining the sequencing process must mitigate errors and ensure robust data to support clinical decisions that are reliable. While sequencing costs are declining, the substantial increase in the cost of the computational analyses needed to process, store, and extract the sequences belonging to the targeted pathogen may be overlooked. Therefore, it is likely that clinical applications of metagenomics will continue to require the use of target-enrichment protocols in order to have a meaningful impact in the clinical practice.

Studies of the placental microbiome are a powerful example of how low-biomass samples can affect results from metagenomics studies. Studies have suggested that the placenta has a distinguishable microbiome^{348–350}. However, the placenta has also been described as a nearly sterile organ with a low biomass concentration, and samples extracted from the basal plate can easily be influenced by contaminants³³⁰ from reagents^{330,351} or the surrounding environment, such as the maternal skin³⁵², which has led to speculation regarding whether it possesses a unique microbiome. Ultimately, studies presented evidence that what was previously interpreted as a placental microbiome was instead likely contamination^{353–356}.

Several optimizations exist to improve the resolution of low-biomass microbial communities and minimize contamination in these samples, such as the use of sterile protocols and appropriate negative controls, as outlined above. Given that reagents can be a source of contamination from either manufacturing or laboratory contamination, these should be treated with DNase as a preliminary step to reduce contamination of low-biomass DNA samples prior to DNA amplification³⁵¹. In the case of the placenta, noted above, specific sterile procedures might include performing thorough disinfection of the maternal skin and using a sterile drape to cover the patient from armpits to knees before and after cesarean³⁵⁷.

Constraints in metagenomic assembly. Challenges in *de novo* metagenome assembly often arise because of the sheer complexity of microbial communities. Metagenomic samples frequently contain a mixture of microbes with different abundances and genomic compositions, including microbes that are genomically difficult to distinguish. Such samples might contain microbial genomes with vastly different coverage depth³⁵⁸ and a complete assembly might only be possible for a few high-abundance organisms.

A high-quality metagenomic assembly is dependent on several features of the sequencing platforms and associated protocols that are used, such as read length, throughput and error rate. At low coverage depths (typically below 3–5-fold), many segments of microbial genomes are not sampled by reads owing to the random nature of the sequencing process. Consequently, the assembly software can only reconstruct a highly fragmented approximation of microbial genomes, with sequence continuity disrupted in regions where coverage is low or absent.

Read length impacts the quality of the sequence assembly owing to the ambiguity caused by repetitive sequences. The presence of repeats hinders the reconstruction of microbial genomes as the assembler might not be able to determine the correct genomic placement of reads contained within repetitive regions. This issue can be addressed by using longer read lengths, which reduce the probability that reads lie completely within or greatly overlap repeat regions³⁵⁹. Paired-end reads are also often preferred over single-end reads as they can help to bridge gaps in assembly (a process called scaffolding) or resolve ambiguity introduced by repeats³⁶⁰.

Metagenomic assemblers must contend with the uneven depth of coverage of contigs within samples (an artifact due to the varied abundances of organisms in a sample), a factor that complicates the detection of repeats and is further confounded by segments of DNA that are shared between different organisms (instead of just being repeated within a single genome)^{113,117}. To account for the effects of sequencing errors and strain-level variation without exacerbating the impact of repeats, metagenomic assemblers tend to integrate assembly graphs generated with multiple *k*-mer sizes. Shorter *k*-mers are more able to overcome sequencing errors or strain variation, whereas longer *k*-mers are more effective at resolving repeats. Some assemblers (such as MetaSpades¹¹⁷) aggressively attempt to “smooth” out sequencing errors and strains in order to generate longer contiguous segments (contigs), while others (such as MEGAHIT¹¹⁴) are more conservative, resulting in more fragmented assemblies.

Long read data can overcome some of the challenges posed by repeats and even strain variants, although it is associated with higher error rates than short read data. Like short read metagenomic assemblers, long read metagenomic assemblers are often developed upon tools initially used to assemble single genomes. Hi-C technology has also been successfully adapted for metagenomics owing to its ability to link together genomic contigs from individual organisms, helping to mitigate the issue of genomic fragmentation typically seen in metagenomic data^{86,361–364}.

Challenges in taxonomic and functional analysis. Challenges associated with taxonomic and functional profiling from metagenomic data include issues regarding accurate classification amid database biases, as well as the complexities of predicting functions and metabolic interactions within diverse microbial communities. Overcoming these hurdles will require interdisciplinary collaboration, advancing computational tools and refining analytical methods.

Taxonomic profiling is inherently a reference-based analysis, where the accuracy of a predicted profile depends on the quality and comprehensiveness of the reference database used. Incorrect selection of reference genomes can result in the exclusion of entire taxonomic kingdoms³⁶⁵ and incorrect taxonomic assignment¹⁵. Differences in nomenclature between taxonomic databases can also alter the distribution of reported taxa. For example, the International Code of Nomenclature of Prokaryotes (ICNP)³⁶⁶ nomenclature integrates genomic information and observed physical characteristics of microbes, whereas the Genome Taxonomy Database (GTDB)³⁶⁷ nomenclature is exclusively reliant on genome phylogeny³⁶⁸. Additionally, it is important to acknowledge that reference

databases are biased towards extensively researched organisms³⁶⁹. Horizontal gene transfer [G] can further blur taxonomic profiling as genes move between microbial genomes. In Supplementary Box 10, we provide an example of how genome database-related biases affect metagenomic analysis, and the transition from taxonomic to functional profiling to improve strain-level resolution.

Outlook

The complexity, multidisciplinary nature, and conceptual and technical challenges of metagenomic research have led to the need for large-scale metagenomics initiatives (Table 3, Supplementary Table 3). Such efforts were established for setting advanced standards in data collection, storage, and sharing, while also assessing the impact of these initiatives on principles and values such as trust, confidentiality and privacy. Large-scale metagenomic initiatives enable the creation of community data resources and common standards while fostering technological advancements, including the development of new tools and shared data and software resources. Consequently, large-scale collaborative metagenomic initiatives have been critical for enhancing our understanding of global biological diversity of environments and human health. Examples of such initiatives include those focused on unmanaged landscapes and aquatic environments, such as Tara Oceans⁴ and Global Ocean Sampling Expedition³⁷⁰, which explored the biological diversity in seawater and sediments. In parallel, the TerraGenome³⁷¹ project explored microbial diversity in soil. Other initiatives focused on characterizing managed ecosystems and the microbial communities within urban environments, including the Earth Microbiome Project⁷ and the MetaSUB initiative³⁷². Another large area of research is host-associated habitats, where initiatives such as the Human Microbiome Project³⁷³, the Tree Root Microbiome initiative, and the BeeBiome Consortium³⁷⁴ focus on examining microbial interactions with hosts including the influence on host health and their interactions within ecosystems. These projects were characterized by centralized management and meticulous design to ensure the production of high-quality data and rigorous downstream data analysis.

We expect that global metagenomic initiatives will continue to support the establishment of broad foundational principles and innovative technologies and methods, given that such advances are more effectively implemented within the context of multi-institutional and highly replicated studies than in traditional single-investigator projects. Additionally, such large projects provide unique opportunities for public outreach and educational development within the metagenomics field. We note, however, that implementing and executing international and multidisciplinary metagenomics projects demands a high level of collaboration and coordination that extends beyond the possibilities of what individual research groups can typically manage. In particular, due to their international character reach, broad scope, and the involvement of numerous investigators, these initiatives necessitate carefully developed management plans, along with funding that is specifically allocated to enhance communication and promote effective collaboration. Despite these challenges, international and multidisciplinary projects offer an excellent opportunity to train a new generation of young scientists, equipping them with skills necessary for collaborative and large-scale science (Supplementary Box 11).

Optimal insights are often achieved through concerted multi-investigator and multi-disciplinary efforts by integrating various methods, such as sequencing, functional-expression analysis, metabolomics [G] analysis, and deep phylogenetic analysis. Consequently, such initiatives will benefit from the broad expertise of professionals from numerous fields, including geneticists, microbiologists, physicians, bioinformaticians, and computational scientists. By integrating diverse expertise and perspectives, researchers can tackle complex challenges more effectively to drive innovation in metagenomic research. Furthermore, leveraging emerging technologies such as machine learning, holds promise for facilitating both data analysis and interpretation. These tools offer opportunities to extract valuable insights from vast amounts of microbiome data, accelerating discoveries and enabling development of personalized approaches to healthcare. Promoting open science principles and data sharing, such as through the establishment of accessible repositories and platforms for sharing data, methodologies, and findings, can maximize the impact of collaborative efforts, fostering transparency, reproducibility, and community engagement.

Building on data values and ethical principles, large-scale metagenomic projects also increase outreach by inviting the scientists from outside the specific consortium to data exploration and analysis. Notably, the Critical Assessment of Massive Data Analysis (CAMDA) challenges and conferences have fostered community driven data analysis for biological Big Data for two decades^{375,376}. Since 2016, CAMDA has partnered with the International MetaSUB Consortium, featuring extensive data sets from large-scale sampling of mass-transit systems and other public areas across the globe generated during City Sampling Day actions. Thus, the CAMDA community has had the opportunity to explore MetaSUB data in the context of microbiome based sample origin prediction or resistome characterization/AMR prediction resulting in over 20 publications to date. Currently, CAMDA participants are exploring new metagenomics data analysis related areas including gut microbiome-based health assessment.

A critical component of large-scale metagenomic research is the refinement of bioinformatic tools, which require rigorous benchmarking³⁷⁶. One practical approach for benchmarking metagenomics tools is through community-organized challenge-based assessments. A notable example is CAMI³⁷⁷, offering challenges dedicated to benchmarking metagenomic tools designed for various tasks such as genome assembly, taxonomic profiling, and binning. These challenges delve into analyzing accuracy, the runtime, and memory usage of bioinformatics tools, providing invaluable insights into their performance. The results obtained from such challenges are instrumental in identifying current limitations of existing computational methods, thereby paving the way for advancements in computational metagenomics.

The range of applications for metagenomics is continually expanding, with profound implications for public health, environmental management, and biodiversity conservation. For example, metagenomics applications have significant potential in the clinical setting, and their implementation is revolutionizing infectious disease diagnostics and pathogen surveillance³⁷⁸. That is, the increased use of metagenomics in patient-care context will allow clinical microbiology to move past simple pathogen identification towards a holistic diagnosis that includes a systematic breakdown of antibiotic resistance, virulence factors

[G], host metabolic determinants, and ecological context. Moreover, sequencing and sequence databases can now be leveraged to learn more about each pathogen diagnosis in the context of the patient, hospital, and the broader community. This ability will allow us to better understand the specific context of each infection, thus facilitating personalized medicine while also allowing epidemiologists to track infection trends in real time. Lastly metagenomics can broadly enhance our understanding of microbial contributions to biodiversity, help us preserve microbial diversity, and assists in identifying rare and unique microbial taxa that will be important for both bioprospecting and the development of new sustainable biotechnological applications to drive future scientific innovations.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Glossary

Viral Quasispecies

A group of closely related viral variants resulting from high mutation rates during viral replication forming a dynamic population of genetically diverse but related sequences, often referred to as a "cloud" of mutants.

Alpha diversity indices

An index that measures species diversity within a single, local microbial community or sample and considers both the number of different species (richness) and evenness of their distribution, as assessed through indices such as Shannon or Simpson's index.

Resistome

The landscape of genes present in the microbiome that are resistant to antibiotic treatment; antibiotic resistant genes can be acquired through horizontal gene transfers, which is an evolutionary event where genes can be moved and adopted among a microbial community.

One-health approach

A metagenomic method that connects metagenomic information of patients, animals, and the environment to encompass and characterize the interaction of microbial communities to aid in the clinical diagnosis and treatment of humans.

Metagenome-Wide Association Studies (MWAS)

A statistical metagenome-wide study that involves the identification and association of genetic loci and genomic information with disease in relation to the host and its environment.

Functional Potential

The presence of genomic elements (such as genes) from which the physiological processes of a microbe can be inferred, but not verified without direct experimental evidence.

Marker Genes

Evolutionarily conserved genes with one or more variable regions that are used as an evolutionary clock to delineate phylogenetic lineages and classify microbes into taxa.

Microbiome

A microbial community sampled from an environment where different species of fungi, bacteria, archaea and viruses can be present; the definition includes microorganisms and their functions and interactions.

Amplicon sequencing

A procedure in which scientists target, amplify, and sequence marker genes, typically the 16S and/or 18S genes ribosomal RNA genes.

Metabolomics

Experimental and computational approaches used to characterize the metabolite profiles found in microbial communities.

Horizontal gene transfer

The exchange of genetic material between organisms that are not in a parent–offspring relationship.

Virulence factor

The molecules that assist microbial pathogens to colonize a host at the cellular level.

Antimicrobial resistance

AMR; the developed capability of bacteria and fungi to resist drugs designed to kill them.

***k*-mers**

A sub-string, of length k , derived from a longer DNA or RNA sequence, used in bioinformatics for tasks such as assembling genomes, analyzing sequence composition, and identifying sequence similarities.

Taxonomic profile

A list describing the taxa in a microbial community along with their relative abundances.

Functional profile

A characterization of the potential biological functions and metabolic pathways within a microbial community.

Functional metagenomics

The study of the genomic roles and interactions within a microbial community; this includes tasks such as gene prediction, functional annotation, and functional profiling to characterize its genomic composition and metabolic and enzymatic activities.

Environmental metagenomics

The study of the microbial community present in a natural ecosystem such as water, soil or air.

Metagenomic assembly

The process of reconstructing individual genomes or genome fragments (contigs) from the sampled DNA of a microbial community.

Metagenome-assembled genomes

MAGs; a genome or collection of genome fragments originating from a single organism extracted from a microbial community.

Metagenomic binning

The grouping of contigs into discrete bins or collections that represent individual organisms or taxa.

Metagenomic profile

A comprehensive overview of collective microbial genetic material from a sample, providing insights into species composition, functional potential, and relative abundance.

De Bruijn graph

Data structure used in genomics to represent overlaps between sequences, where nodes represent k -mers (substrings of length k) and directed edges represent overlaps of $k-1$ bases between consecutive k -mers; the sequence of the genome being assembled can be “read” by traversing the graph and concatenating the sequence of the k -mers encountered.

High-performance computing (HPC) clusters

A network of computers that work in tandem to efficiently tackle intensive computational tasks.

Sequencing depth

The number of times that a particular DNA or RNA nucleotide is read during the sequencing process. The average depth of sequencing coverage can be defined as LN/G , where L is the read length, N is the number of reads, and G is the haploid genome length.

Antibiotic Resistance Genes

ARGs; specific genetic sequences within microbial communities that encode factors conferring resistance to antibiotics; these genes are often located on plasmids or transposons and can be transferred from cell to cell by conjugation, transformation, or transduction.

Shotgun metagenomics

An approach that enables the analysis of an entire microbial community without targeting specific species by randomly fragmenting the genomic material present in a metagenomic sample and sequencing it.

Open Reading Frame

A continuous sequence of codons in a genomic region, starting with a start codon and ending with a stop codon, that has the potential to be translated into a functional protein.

Environmental DNA

Genetic material obtained directly from environmental samples such as soil, water, and air.

Contigs

Contiguous sequences of DNA that are assembled from overlapping sequence reads.

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Box 1 |**Environmental metagenomics applications****Food safety metagenomics.**

Metagenomic studies have profiled the microbial communities present across various stages of the food production and supply chain industries. Samples include food products, ingredients, and environmental samples such as processing plant surfaces³⁷⁹, storage temperatures³⁸⁰ packaging³⁸¹, irrigation water³⁸², and even additives such as salt³⁸³. Leveraging metagenomic techniques can advance our understanding of microbial interactions within the food industry and enable more refined protocols for the early detection and prevention of foodborne pathogens³⁸⁴.

Soil metagenomics.

Metagenomics research has illuminated microbial diversity across various soil types and highlighted dramatic alterations in soil microbiome composition over time due to industrialization and increasing contamination. For example, metagenomics analyses have revealed the complex interactions between pesticides and microbial genes involved in pesticide degradation, underscoring the potential of soil microbial communities to mitigate pesticide contaminations³⁸⁵. They have also aided in discovering beneficial microbes that promote plant growth and protect against diseases, contributing to more resilient agricultural systems^{284,386}. In addition to soil-based studies, metagenomic research has been applied to farm animals to track viral pathogens and determine the relationship between the microbial community and host nutrition and metabolism³⁸⁷.

Water metagenomics.

Water ecosystems including oceans⁴⁵, lakes^{388–390} (including artificial³⁹¹ and volcanic lakes³⁹²), rivers³⁹³ and mangroves³⁹⁴ have been extensively studied through metagenomics. Functional metagenomics[**G**] analyses of deep ocean sources have revealed a surprising diversity of metabolic strategies among microorganisms⁴⁵. Results from metagenomic analyses of wastewater and contaminated groundwater provide crucial insights for developing effective recycling and bioremediation methods³⁹⁵. Metagenomic tools have also enhanced the assessment of microbial communities in drinking water for water quality monitoring³⁹⁶. These methods can track indicators of fecal pollution, detect common bacteria in polluted water, and explore the presence of viruses.

Air metagenomics.

Metagenomic studies of air samples are challenging due to low microbial density and the lack of standardized methodologies³⁴⁴. However, these studies revealed a positive correlation between certain airborne microbial genera and mortality rates in patients with respiratory diseases and identified positive and negative correlations between anthropogenic activities and airborne microbial communities³⁹⁷. Hospital air directly impacts patient health and is a crucial focus for metagenomic research^{398–400}; shotgun metagenomic approaches have identified an abundance of opportunistic pathogens in hospital air^{264,399}.

Metagenomic bioengineering.

The integration of synthetic biology with functional metagenomics has revolutionized the discovery and exploitation of novel genes and biochemical structures from metagenomic profiles. This synergy has led to the identification of unique enzymes, and molecular interactions with therapeutic potential. For example, the interaction of N-acyl-amide synthases and G-protein-coupled receptors⁴⁰¹, and the discovery of sialidases within the glycoside hydrolases family (GH156)⁴⁰² in gastrointestinal bacteria underscore the potential of metagenomic bioengineering to reveal previously inaccessible molecular targets. The application of synthetic biology on marine metagenomics has enabled the reconstruction of complete genomes of previously unknown microorganisms, leading to the discovery of new genes and metabolic pathways with significant biotechnological potential⁴⁰³. Further, metagenomics serves as a valuable resource for mining new enzymes from microbial community metabolomes through functional screening and sequence-based approaches⁴⁰⁴. A wide range of enzymes, including lipases, cellulases, and proteases, have been discovered, with metagenomic analyses promising even more discoveries in the future⁴⁰⁵.

Non-human microbiome.

Profiling non-human species has been essential in comparative metagenomics. For example, a metagenomic study on the gut microbiomes of various non-human primate cohorts revealed an array of previously unidentified microbes and a 20% shared composition of species with the human microbiome⁴⁰⁶. Further, the use of metagenomic data from non-human species can shed light on how microbial communities co-evolve between and within species. A metagenomic study across three species of carpenter bees evaluating the core microbiome among these species revealed genomic variations⁴⁰⁷.

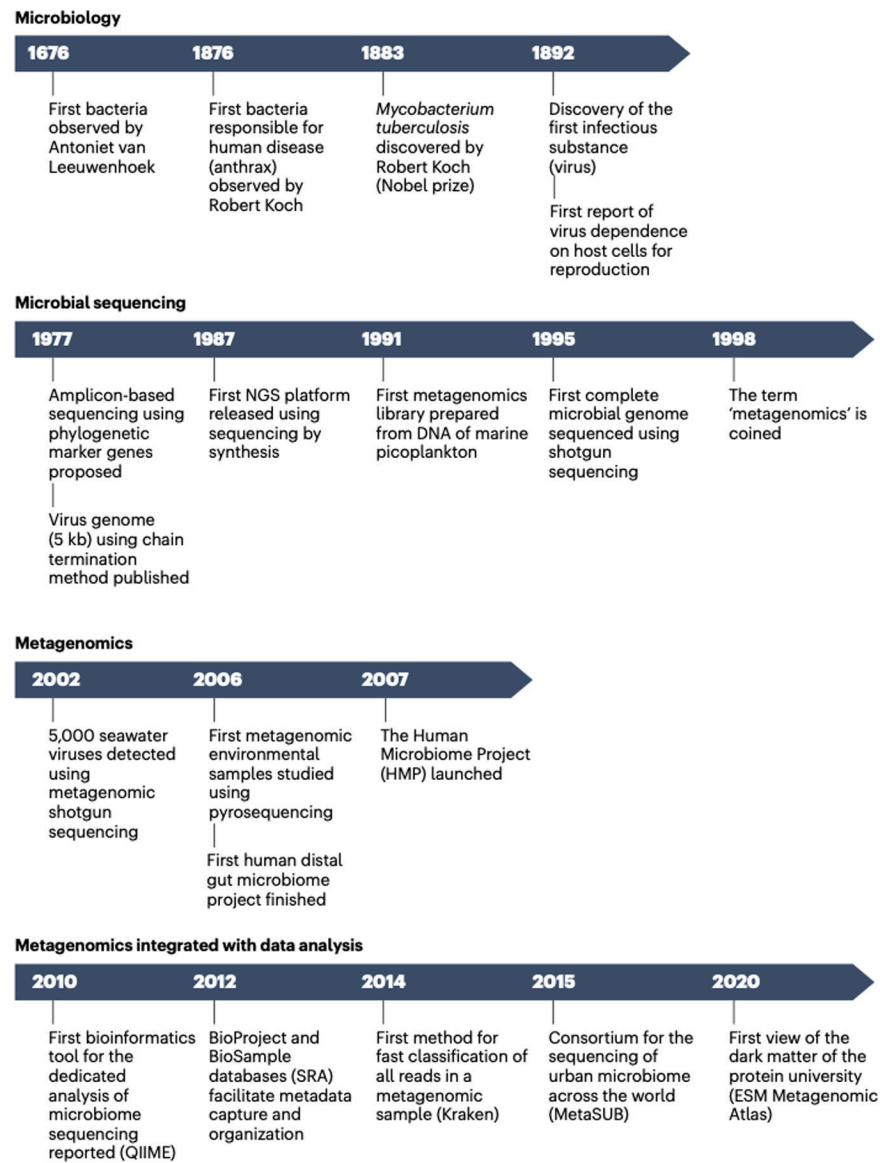


Figure 1. A timeline of microbial discovery and the development of metagenomic analysis. From the findings of early microbiology and advancements in sequencing technology, the field of metagenomics has grown to encompass the development of metagenomic databases, tools, and organizations dedicated to the field of microbial and metagenomic discovery.

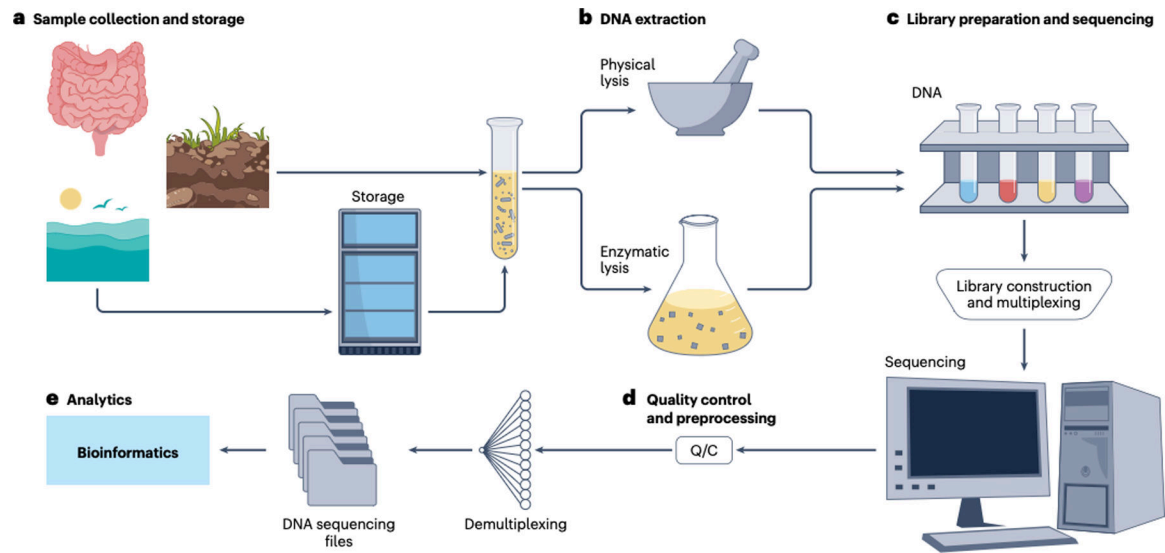


Figure 2. Experimental protocol for metagenomics experiments.

a, Metagenomic samples, such as environmental samples from soil or water, or samples from the microbiome of organisms, are collected and either stored or processed immediately. b, DNA from the sample is extracted using physical or enzymatic lysis. c, A DNA library is constructed. Multiple samples can be sequenced together (multiplexed) by labelling samples with DNA barcodes. Bulk DNA is sequenced using whole shotgun sequencing. d, Sequencing reads undergo quality control checks and preprocessing. e, Demultiplexing separates sequences by DNA barcode and sequences are then processed using bioinformatics analyses.

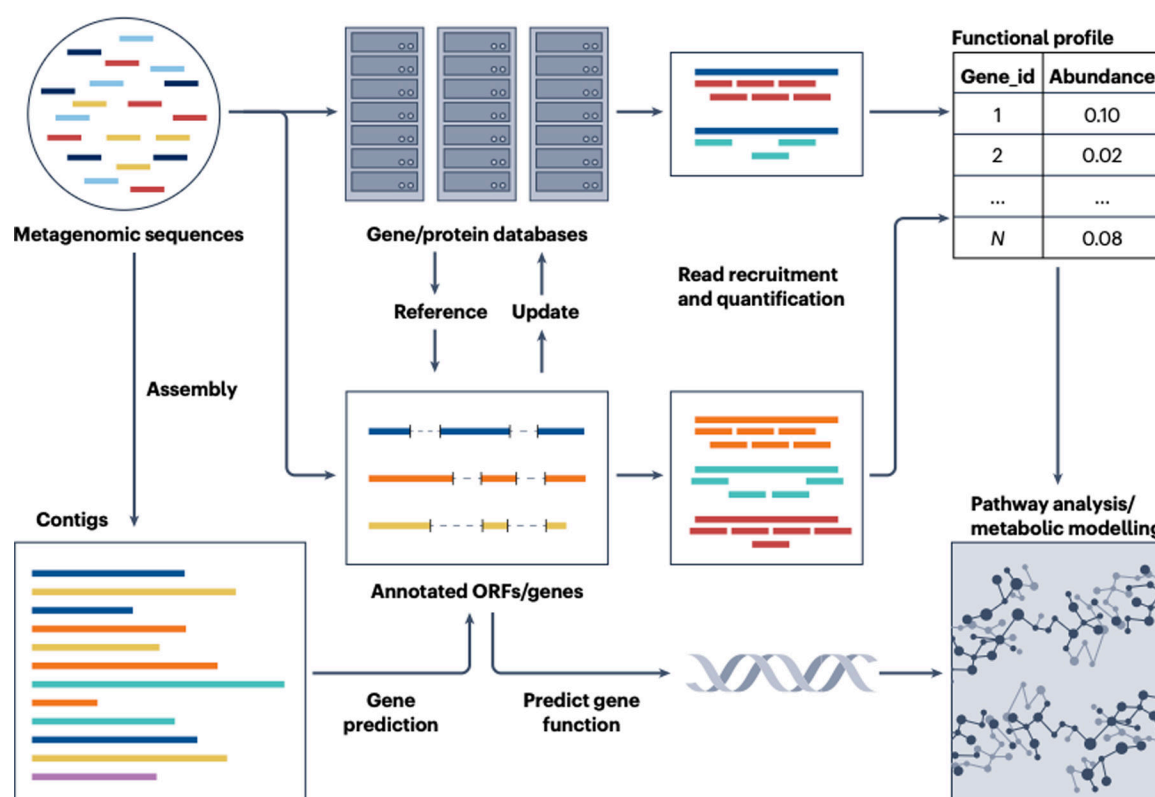


Figure 3. Metagenomic functional analysis.

Metagenomic sequences are a common starting point for functional profiling and annotation. Known genes can be directly aligned to a reference for alignment-based quantification (top row). Metagenomic sequencing reads can also be employed by *ab initio* gene finders to train models that identify ORFs from input data followed by reads quantification (middle row). Both alignment-based and *ab initio* methods can be used for functional profiling, which can be further analyzed to explore biological insights (right side). Metagenome sequences can be assembled into contigs to facilitate gene discoveries by identifying novel ORFs and predicting gene function based on existing knowledge (bottom row).

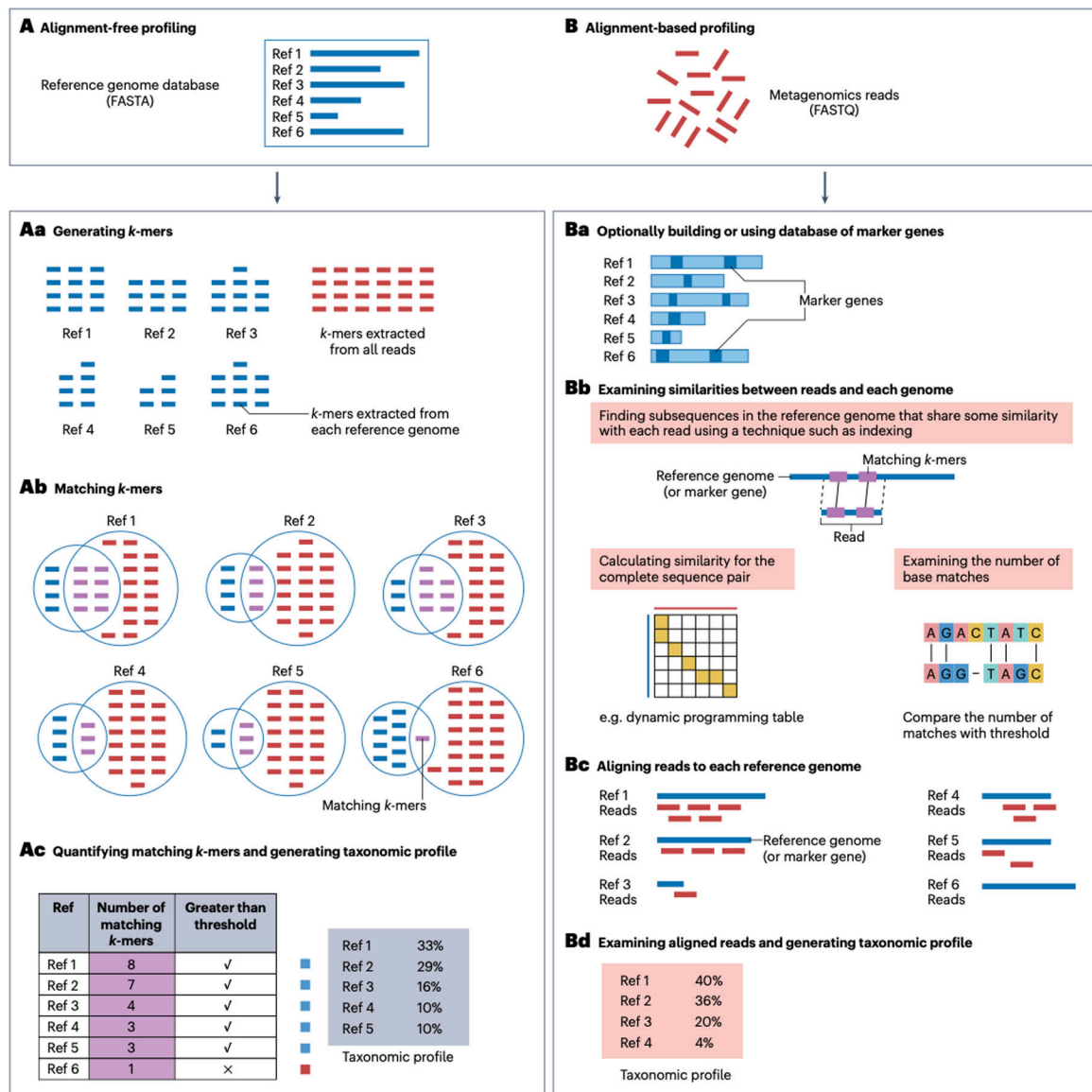


Figure 4. Metagenomic taxonomic characterization using alignment-based and alignment-free methods.

In both alignment-based and alignment-free profiling approaches, a public reference database is chosen for the input of the metagenomic data. **A**, Alignment-free profiling tools in general experience reduced computational burden. **Aa**, The generation of *k*-mers is performed on both the chosen reference dataset and the metagenomic sample of interest. **Ab**, *k*-mers generated from the references and sample are compared to identify matching *k*-mers. **Ac**, Matching *k*-mers are quantified to generate a taxonomic profile. **B**, Alignment-based methods can be more sensitive at the cost of increased computational burden. **Ba**, The metagenomic sequencing reads are aligned to a prebuilt marker-gene-based reference database. **Bb**, Similarities between reads and each genome are calculated using either indexing, dynamic programming, or *k*-mer matching. **Bc**, Reads are then aligned to the reference database to be quantified for taxonomic profiling (**Bd**).

Table 1.

Examples of publicly available databases that store microbial reference genomes.

Database	Total species (approximately)	Contents
RefSeq	125,000	Comprehensive, non-redundant, well-annotated set of reference genomic sequences presented in GenBank database, including archaea, bacteria, fungi, and viruses
Ensembl ⁴⁰⁸	300,000	Vertebrate genomes, plants, fungi, bacteria, and protists.
VEuPathDB ⁴⁰⁹	500	Eukaryotic pathogens (protists and fungi) and relevant free-living or non-pathogenic species.
BV-BRC ^{*302}	87,000	Combines the PATHosystems Resource Integration Center (PATRIC), the Influenza Research Database (IRD) and the Virus Pathogen Database and Analysis Resource (ViPR) and includes bacterial and viral pathogens, and archaeal genomes.
Unified Human Gastrointestinal Genome (UHGG) ⁴¹⁰	204,000	Non-redundant genomes for human gut prokaryotes
JGI's 1000 fungal genomes project (JGI 1K) ^{*300}	388	Fungal and algal genomes
Human Reference Gut Microbiome (HRGM) ^{*304}	232,000	Non-redundant genomes representative prokaryotic species
Microbial Genome Database (MBGD) ^{*411}	15,000	Incorporates all complete genome sequences of bacteria, archaea, and unicellular eukaryotes including fungi and protozoa available at the NCBI genomes FTP site.

Links to access databases and FTP servers are provided in Supplementary Table 4.

* Hyperlink for database does not link to an FTP Server.

Table 2.
Most-used publicly available repositories for metagenomic data

Database	Total metagenomic samples (approximate)	Description
ENA/SRA	2,000,000	Largest public repository of high-throughput metagenomic data, with cloud hosting for fast computation without downloads.
MGNfy	343,000	Platform for harmonized assembly, functional and taxonomic analysis of diverse meta-datasets including sample storage and inter-study associations.
MG-RAST *	512,000	Upon upload, the pipeline performs QC, functional, and phylogenetic analysis. Real and simulated data can be stored as public or private.
Tara Oceans	35,000	Database focusing on Sunlit ocean life, consisting of seawater and plankton samples from 210 ocean stations. Includes data from the Global Ocean Sampling expedition, Pacific Ocean Virome project, NCBI reference genomes, and Moore Microbial Genome Sequencing Project.

Sample number is estimated without counting amplicons.

* MG-RAST is currently not actively supported by any organization.

Table 3.

Details of publicly available initiatives dedicated to the standardization of metagenomic data.

Initiative	Total metagenomic samples (approximate)	Description
HumanMetagenomeDB 1.0	69,000	Standardizes human metagenomic data. Sequencing data isn't available on the website, but a script is provided for downloads.
TerrestrialMetagenomeDB 2.0	20,000	Focused on the standardization of metadata for terrestrial metagenomes. Similar to HumanMetagenomeDB 1.0; developers offer a download script.
PlanetMicrobe	2,000	Specializes in the standardization of marine or non-marine aquatic samples.
MetaSUB Consortium	20,000	Standardization of collection, storage, transport, processing and analysis of surface, air, water and sewage samples from urban and rural areas.
Serratus Database	5,700,000	Access to raw virome sequencing data including RefSeq vertebrate viruses, GenBank, Coronaviridae, and full-length RNA-dependent RNA polymerase sequences, standardizing the processing of contigs including assembly and annotation
National Microbiome Data Consortium (NMDC)	2,900	Integrates the standards of multiple organizations to evaluate quality and access for metagenomic data, in addition to other multi-omics data such as proteomics, metatranscriptomics, and metabolomics.
Critical Assessment of Metagenome Interpretation (CAMI)	526	Comprehensive evaluation of metagenomic tools, offering reliable performance data to guide accurate interpretation and method improvement